JYX
JYVÄSKYLÄN YLIOPISTO
UNIVERSITY OF JYVÄSKYLÄ

This is a self-archived version of an original article. This version may differ from the original in pagination and typographic details.

Author(s): Ghanbari, Hadi; Tuunanen, Tuure; Kazan, Erol

Title: Towards competition and collaboration : a mid-range design theory for designing coopetitive news platforms

Year: 2026

Version: Published version

Copyright: © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Franci

Rights: CC BY 4.0

Rights url: https://creativecommons.org/licenses/by/4.0/

Please cite the original version:

Ghanbari, H., Tuunanen, T., & Kazan, E. (2026). Towards competition and collaboration : a mid-range design theory for designing coopetitive news platforms. European Journal of Information Systems, 35(1), 26-53. https://doi.org/10.1080/0960085X.2025.2525371

The University of Jyväskylä's Rights Retention Strategy (RRS) enables accepted versions (final drafts) to be made openly available through self-archiving. For more information, see: https://www.jyu.fi/en/rrs



European Journal of Information Systems

ISSN: 0960-085X (Print) 1476-9344 (Online) Journal homepage: www.tandfonline.com/journals/tjis20

Towards competition and collaboration: a mid-range design theory for designing coopetitive news platforms

Hadi Ghanbari, Tuure Tuunanen & Erol Kazan

To cite this article: Hadi Ghanbari, Tuure Tuunanen & Erol Kazan (09 Jul 2025): Towards competition and collaboration: a mid-range design theory for designing coopetitive news platforms, European Journal of Information Systems, DOI: 10.1080/0960085X.2025.2525371

To link to this article: https://doi.org/10.1080/0960085X.2025.2525371

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

View supplementary material

Published online: 09 Jul 2025.

Submit your article to this journal

Article views: 820

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tjis20

EUROPEAN JOURNAL OF INFORMATION SYSTEMS
https://doi.org/10.1080/0960085X.2025.2525371

RESEARCH ARTICLE

OPEN ACCESS Check for updates

Towards competition and collaboration: a mid-range design theory for designing coopetitive news platforms

Hadi Ghanbari[a], Tuure Tuunanen[b] and Erol Kazan[c]

[a]Department of Information and Service Management, Aalto University, Espoo, Finland; [b]Faculty of Information Technology, University of Jyväskylä, Jyväskylä, Finland; [c]Deutsche Bundesbank, Frankfurt am Main, Germany

ABSTRACT

Newspapers' digital transformation offers various opportunities and challenges for publishers. While some publishers, such as *The New York Times*, have been able to monetize their online content, many publishers struggle to sustain revenues. Drawing on the strategic alliance literature, we propose that a coopetitive news platform co-owned and operated by multiple competing publishers for cross-selling online news could resolve this problem. However, we lack theoretically grounded design knowledge for designing such a coopetitive news platform. We conducted a 5-year echeloned design science research ([e]DSR) project to address this gap. Through 16 design echelons, we developed a news platform prototype and produced design knowledge for developing it. Our findings offer proof of concept and proof of value for the proposed artefact from users' and online news publishers' perspectives. Our contributions to the literature and practice are several. We developed design principles and abductively built a mid-range design theory for designing coopetitive news platforms that loosely couple platform proxies, enabling them to form a coopetition alliance. This, in turn, creates novel opportunities for bundling and cross-selling content from various competing publishers via novel coopetitive subscription plans.

ARTICLE HISTORY
Received 16 January 2024
Accepted 20 June 2025

KEYWORDS
Coopetitive news platform; online news; digital subscription; design principles; mid-range design theory; echeloned design science research

1. Introduction

Newspapers are essential for facilitating a public discourse to inform citizens and hold stakeholders accountable. To fulfil their roles in the digital age, publishers have adopted digital channels to serve online readers. Thanks to ubiquitous computing on different devices, online channels have gained considerable importance for most publishers in monetizing their online content (Pattabhiramaiah et al., 2019). However, these revenues have become increasingly unpredictable. Given the prevalence of online ad blockers, the rise of privacy-aware users, data-protective computing platforms (e.g., Apple), and regulations (e.g., the European ePrivacy directive 2009/136/EC) that hinder advertisers from tracking users as well as competition from public broadcasters and news aggregator platforms (e.g., Google News, Apple News+), newspaper publishers are struggling to create steady and consistent revenues from their online content.

Responding to these market dynamics, publishers have increasingly opted to guard their content through paywalls, which require digital subscriptions. Although digital subscriptions promise stable revenues, and large publishers with a global readership have indeed benefited (e.g., *The New York Times*), smaller publishers are challenged to replicate similar outcomes. Local newspapers struggle to convince readers to pay for digital subscriptions in the first place (Chyi & Tenenboim, 2019; Sjøvaag & Owren, 2021). While readers have discretionary media spending, similar content is freely available just a few clicks away. Furthermore, even if readers take out digital subscriptions, the churn rate (i.e., the cancellation rate) is high, suggesting a value misalignment between users, price, and news content (Olsen & Solvoll, 2018). Global platforms have recognized the problem and started to offer news aggregators that pool content from different publishers in one place (e.g., Apple News+) with a market-orientated subscription price. Recent research has shown that inclusion in news aggregators can increase an online newspaper's website traffic and revenues (Calzada & Gil, 2020), thus creating a net market expansion. From the reader's perspective, this may arguably address the value misalignment pertinent to content attractiveness. Conversely, many publishers are sceptical about joining these aggregators due to concerns about being commoditized by new intermediaries, which implies a loss of control over distribution and user relationships and ultimately deprives already financially constrained publishers of their monetization options (Lee, 2019).

Considering the prevailing trends in most media industries, such as music (e.g., Spotify), books (e.g.,

CONTACT Tuure Tuunanen tuure.tuunanen@jyu.fi
Supplemental data for this article can be accessed online at https://doi.org/10.1080/0960085X.2025.2525371

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Amazon Kindle), movies (e.g., Netflix), and now newspapers (e.g., Apple News+), questions arise regarding how to design online newspaper platforms that address a significant concern of publishers: dependence on a single actor with monopolistic power (Lee, 2019). One solution is to share risks and opportunities collaboratively through cross-selling (cf. Li et al., 2011) online news content. A collaboratively operated digital news aggregator platform, hereafter called a *news platform*, could alleviate some of the publishers' concerns and offer added revenue streams to newspapers without cannibalizing their current customer base. In Europe, we have already seen such initiatives. For example, the Austrian News Agency (APA), a co-operative agency owned by Austrian newspapers and the national broadcaster ORF, developed a platform for facilitating collaboration among newspapers (Jääskeläinen et al., 2021) and media professionals to share news content.

Such organizational arrangements are similar to strategic alliances (see, e.g., Ryan-Charleton et al., 2022) successfully implemented earlier in digital industries, such as fintech (Klus et al., 2019). Although such collaborative ventures present an opportunity to share news content, leverage joint digital technology and possibly create additional revenue sources, we lack knowledge on how to design such a collaborative but also competitive platform. This form of strategic alliance is called *coopetition*, in which competitors simultaneously cooperate for common benefits (i.e., value creation) and compete for private benefits (i.e., value capture) (Gulati et al., 2012; Nalebuff & Brandenburger, 1997; Stadtler & Van Wassenhove, 2016; Yadav et al., 2022). This type of arrangement is quite different from other organizational collaborations, such as R&D joint ventures in which services or products are vertical complements (e.g., a car manufacturer and a supplier), thus serving different customers and markets. A strategic alliance is also different from business concerns like Finnish media groups Mediatalo Keskisuomalainen or Sanoma Media Finland, which offer news in a number of outlets under one management. A news platform co-owned and operated by multiple competing publishers warrants different collaboration dynamics, business models, and technology compared to existing and monopolistic services. It requires a new conceptualization of how newspaper publishers operate and, more importantly, can collaborate but also compete within the same news platform (see, e.g., Jääskeläinen et al., 2021; Piepponen et al., 2022). Prior research has provided valuable insights into different commercialization efforts in the newspaper industry, including paywalls, freemium business models, and news aggregators (Berger, 2018; Calzada & Gil, 2020; Koch & Benlian, 2017; Pattabhiramaiah et al., 2019). However, a collaborative news platform that would enable the cross-selling of online news content is largely unexplored. Thus, we argue that many reasons exist for competing newspaper publishers to cooperate and offer joint digital subscriptions to customers, including using fewer resources and avoiding duplication of effort (Brandenburger & Nalebuff, 2021).

Value creation and capture have been the focal areas of research in the coopetion literature (see, e.g., a recent literature review by Corbo et al., 2023). However, while researchers have argued that this kind of coopetion can lead to business model innovations (Chiambaretto et al., 2020; Kraus et al., 2017; Ritala, 2012), the literature also highlights that coopetition is a rare organizational constellation, as production (i.e., value creation) and competition in markets (i.e., value capture) are difficult to align in a business model (Geurts et al., 2022). Consequently, it is not surprising that only a small subset of the coopetition literature has examined the effects of cross-selling products and services (see, e.g., Luo et al., 2007). Most studies on coopetition and cross-selling have focused on how this impacts a firm's performance when it implements coopetition and cross-selling within the firm (see, e.g., Malms & Schmitz, 2011). Importantly, the literature does not offer guidance on how to design platforms that support the coopetitive cross-selling of products or services.

We conducted an echeloned design science research ([e]DSR) study (Tuunanen et al., 2024) seeking to generate design knowledge by formulating and evaluating a set of design principles (DPs) for designing coopetitive online news platforms to answer the following research question: What are the design principles of a coopetitive digital platform that support coopetitive cross-selling of online news content?

Adhering to [e]DSR, the news platform (i.e., the artefact) was demonstrated in academic and industry workshops, refined based on the feedback and evaluated with a free simulation experiment ($n = 175$) and a concept testing survey ($n = 415$) to evaluate users' experiences and subscription plan choice. This study contributes to the coopetition literature in management and online newspaper literature in information systems (IS) by unpacking the elements, dimensions, and dynamics of designing a coopetive news platform. Our study also contributes to the design science research (DSR) literature by presenting how we abductively build a mid-range design theory[1] (Gregor & Hevner, 2013; Merton, 1968) and how the related DPs evolved during the project as well as by showcasing a template of how to organize, conduct, and report an [e]DSR project. This study also has significant implications for the newspaper industry by presenting a way to solicit tension points and derive the positive conditions needed for collaboration and

increasing digital subscriptions and, thus, revenues without cannibalizing their existing customer base.

2. Theoretical background

2.1. Business models for online newspapers

The newspaper industry has witnessed considerable digital transformations, presenting a rich empirical context to study digitalisation from different theoretical and analytical perspectives. Recent studies have explored the open innovation capabilities of Danish and Swedish publishers (Holm et al., 2013; Thorén et al., 2014), unpacked newspaper price dynamics when publishers merge (Chandra & Collard-Wexler, 2009), examined how online newspapers attract different stakeholders (Evans, 2009), explored how publishers leverage dynamic capabilities during industry transformation (Karimi & Walter, 2016) and how online publishers become entrepreneurial as they experiment with novel delivery channels and business models (Fetscherin & Knolmayer, 2004; Karimi & Walter, 2016). Regarding business models, studies have, for example, explored how credibility impacts a publisher's monetization efforts (Berger, 2018), how mobile apps support reducing churn rates (Winter & Alpar, 2018) and unpacked the conversion rate between freemium business models (Koch & Benlian, 2017) and the relationship between free and paid news articles (Fetscherin & Knolmayer, 2004; Sjøvaag, 2016).

Research has paid considerable attention to business models to explain the logic of how organizations create and capture value. Timmers (1998) work of the late 1990s was highly impactful for understanding many aspects of early digital business models, such as e-shops, e-procurement, e-malls, third-party marketplaces, and information brokers. Amit and Zott (2001) conceptualized value creation and illustrated how different business models generate value. Hedman and Kalling (2003) and Magretta (2002) emphasized the importance of understanding how a business model supports creating and delivering customer value. Osterwalder et al. (2005) seminal work on the business model canvas proposed a framework for depicting the essential elements of a business model. Later, Al-Debei and Avison (2010) proposed four generic dimensions of digital business models: value proposition, value finance, value architecture, and value network. Considering these dimensions in the online newspaper context, publishers rely on their digital infrastructure and online delivery channels (i.e., value architecture) to create and offer digital news content (i.e., value proposition), and in collaboration with internal and external stakeholders (i.e., a value network), to generate revenue (i.e., value finance).

Traditional newspapers have relied heavily on revenue generated from readers and advertisers, in which economies of scale played a key role (Jenkins & Nielsen, 2020; Rios-Rodríguez et al., 2024). In this model, many local newspapers enjoyed a near monopoly (Picard, 2018), being well positioned to provide readers with unique local information and supply potential advertisers with geographical segmentation, an advantage reflected in the concept of non-substitutability (Sjøvaag & Owren, 2021). While this traditional print-based business continues to reliably generate revenue for some news publishers (Jenkins & Nielsen, 2020; Rios-Rodríguez et al., 2024), economic challenges caused by the 2008 global financial crisis (Myllylahti, 2016), the newspaper industry's digitalization and changing reader behaviour have reduced newspaper publishers' profitability (Bhuller et al., 2024; Chyi & Jeong, 2024; Chyi & Tenenboim, 2017; Myllylahti, 2016; Rios-Rodríguez et al., 2024). Moreover, newspapers' advertising revenue has continued to decline (Chyi & Tenenboim, 2017; Jenkins & Nielsen, 2020).

Consequently, local news publishers have attempted to diversify their revenue streams by focusing on digital subscriptions and paywalls and using analytics to offer personalized content (Jenkins & Nielsen, 2020; Myllylahti, 2016; Puijk et al., 2021; Sjøvaag, 2016). Online news publishers can benefit from greater efficiency and economies of scale enabled by digitalization, especially when publishing news only online, thus reducing their materials costs (Rios-Rodríguez et al., 2024). Considering that local news publishers often operate in small markets, they have traditionally faced difficulties in scaling up their subscriber base to an optimal size; therefore, digitalization is of particular interest to them (Rios-Rodríguez et al., 2024). However, research has shown that publishers have difficulties convincing readers to pay for digital subscriptions (Chyi & Tenenboim, 2019; Sjøvaag & Owren, 2021). Olsen and Solvoll's (2018) study of the Norwegian newspaper industry suggested that daily newspapers with paywalled content continued to struggle and posited several value misalignments regarding content, reader perceptions, and price. While older readers were loyal print subscribers, they hardly used the complementary digital channels (Olsen & Solvoll, 2018). Furthermore, younger and low-income readers shunned paywalled local content, tending to consume freely available news from alternative sources.

Other studies have shown that online news' global reach has especially challenged the non-substitutability of local newspapers' offerings, impacting their advertising market control (Sjøvaag & Owren, 2021). Digitalization has also shifted the focus of advertising from large-scale campaigns to direct, targeted access to precise segments, reducing

advertising income (Sjøvaag & Owren, 2021). To deal with these issues and sustain revenue, news publishers have mostly increased their subscription fees or tried to attract potential advertisers by providing user metrics (Sjøvaag & Owren, 2021). However, subscription price increases have been shown to increase the subscription churn rate (Chyi & Tenenboim, 2019).

Considering the literature and current industry developments (e.g., news aggregators), a common theme emerges: Most newspapers have difficulty identifying steady and reliable revenue sources while facing the undesired developments of intermediation by monopolistic third parties. Although newspaper publishers compete fiercely, most collaborate in various ways and degrees to pool their resources (e.g., content licensing) or coordinate to standardize certain aspects of their operations (e.g., user data regulations). Such collaborations, however, are predominantly noncommercial. It is worth exploring how publishers could collaborate on a commercial level that mitigates monopolistic influences but also presents new avenues for generating additional revenue.

2.2. From competition to coopetition

Competition in network economies is increasingly driven by interorganizational arrangements in which multiple organizations pool their respective resources and capabilities to create valued products and services. Such organizational arrangements can be joint ventures (e.g., R&D between organizations), strategic alliances (e.g., Amazon and BMW), or industry consortia (e.g., national payment systems). In the management literature, such collaborations are often referred to as strategic alliances (Gulati et al., 2000; Ryan-Charleton et al., 2022), in which organizations compete in one area, usually on the consumer side, and collaborate in other areas to achieve mutual benefits, such as economies of scale and scope.

Strategic alliances are difficult to create or imitate because they result from complex managerial negotiations and require the disclosure of proprietary knowledge, a key source of firms' competitiveness (Gulati et al., 2000). Similarly, the ability to enter alliances is unevenly distributed, as certain resources can be perceived as tradeable or duplicative substitutes, weakening the position of alliance seekers (Gulati et al., 2000). Recent research has aimed to distinguish between horizontal and vertical alliances (Hoffmann et al., 2018). Horizontal alliance members operate in the same industry and compete for the same customers. Gnyawali and Park (2011) study explored the collaboration dynamics (i.e., a horizontal R&D joint venture) between Samsung and Sony in developing next-generation TV screens. Vertical alliances typically illustrate close or exclusive relationships between focal firms (e.g., manufacturers) and complementary firms (e.g., suppliers) that provide components to the former. The collaboration between the car manufacturer BMW and Amazon exemplifies a vertical alliance, equipping BMW vehicles with virtual voice assistants (i.e., Alexa).

In the newspaper industry, horizontal R&D alliances are uncommon, but often, partnerships have been with technology providers, as in the case of the *Financial Times* entering a vertical alliance with OpenAI to leverage newspaper data for its generative artificial intelligence tools.[2] The European Union has focused on offering financial incentives in the form of different funding programmes for media companies to form horizontal R&D alliances (cf. https://digital-strategy.ec.europa.eu). The Trusted European Media Data space (TEMS) project is one such research initiative, currently focusing on developing a business-to-business exchange platform for fact checking and news content, collaboration in production chains, and a syndication platform for written works that focuses more on how newspapers have already collaborated in their business. Some TEMS trials are also considering content sharing in the form of syndication and using artificial intelligence to tailor the content to consumers. Such syndication is similar to the APA case, where the agency moved towards a platform-based business model to support regional newspapers and stay relevant to the Austrian news market (Jääskeläinen et al., 2021). However, Jääskeläinen et al. (2021, p. 2075) noted that "APA did not use a planned and strict platform-design framework or architecture to set up the platform business model" (i.e., the APA did not purposefully design a news platform to support the new business model). Thus, how different newspapers could collaborate within a news platform while still competing for readers has not been considered.

Such complex horizontal alliances, commonly referred to as *coopetition* (Bouncken et al., 2015; Gulati et al., 2012; Nalebuff & Brandenburger, 1997), are particularly challenging to form and maintain, as organizations are often direct competitors that are reluctant to reveal the sources of their competitive advantages. Horizontal alliances often focus on collaboration, predominantly on the supply side (i.e., production and R&D), to co-develop products. Coopetition's collaborative nature usually ends once products and services reach the market, transitioning towards a competitive mode, as observed between Samsung and Sony (Gnyawali & Park, 2011). While value creation and capture have generally been considered independent outcomes of coopetition, the literature also sees these outcomes as interdependent (Corbo et al., 2023). Meena et al. (2023) also linked value creation to innovation performance, referring to Ritala and Hurmelinna-Laukkanen (2009) as an example of firms' dynamic capability to create value while

collaborating on R&D projects. However, coopetition studies often highlight the fact that coopetition principles (i.e., value creation and capture) are difficult to align in a business model (Geurts et al., 2022).

Nevertheless, there is support in the literature for developing such business models if there are distinct joint industry challenges that could be resolved by entering a coopetition arrangement. Choi et al. (2010) studied the cases of the Australian, New Zealand, and US wine industries, finding that five factors facilitate the formation of coopetition: strategic common goals, greater expected collective benefits, safeguards against knowledge transfer, intra-industry complementarities, and strong ties in cooperative networks. Mathias et al. (2018) reported similar findings in the craft beer industry and found that pulling together and "paying it forward" enticed incumbent and new craft manufacturers to continue their coopetition approaches. In the circular economy setting, Harala et al. (2023) argued that the purposeful design of a platform-based business model is essential for moving forwards from the traditional coopetition form of pre-competitive R&D to sharing a joint news platform for collaboration while also competing for customers. That said, the literature also warns of the challenges in entering into such horizontal coopetitive alliances (Ritala, 2012), and researchers even advise "Keep your friends close, but your competitors closer" (Luo et al., 2007, p. 81). Coopetition can lead to mixed results if tensions caused by coopetition are mismanaged (Le Roy & Czakon, 2016).

For small and medium-sized companies, which are often reluctant to enter such alliances, a coopetitive approach can offer clear benefits to innovate their business model (Chiambaretto et al., 2020). Research shows that "small firms are less reluctant to coopete than large firms, especially if coopetition allows them to reduce their costs and learn from their coopetitor" (Chiambaretto et al., 2020, p. 1). Furthermore, Ritala and Sainio (2014) argue that coopetition can lead to radical innovations in business models that do not always require radical or disruptive technological innovations. While these findings are highly interesting, they do not offer ways to operationalize a coopetitive platform-based business model for cross-selling online news content.

2.3. From traditional to coopetitive digital platforms

Digital platforms are technology architectures (Yoo et al., 2010) that orchestrate transactions between various platform stakeholders to foster innovation (Kazan et al., 2018). In traditional platforms, an organization owns and controls the technology architecture and its components (i.e., value architecture) and captures revenue through transaction fees or subscription fees paid by platform members and consumers for using the platform resources or through advertising fees paid by advertisers. In return, platform complementors or members have the right to leverage the platform's value architecture and value network to develop and deliver services and offerings without intervention from competitors (Kazan et al., 2018; Yoo et al., 2010). In such arrangements, a significant concern is platform members' dependence on a single operator, such as Apple or Amazon, with monopolistic power over the platform (Lee, 2019). In addition, digital platforms are constantly confronted with balancing the needs of existing and new stakeholders with different roles, such as complementors, developers, service providers, consumers, and buyers (i.e., value network), to maintain the platform's attractiveness and competitiveness while avoiding fragmentation that would degrade the overall service quality (Ellis et al., 2003).

A coopetitive or consortium-based platform (Gulati et al., 2000) could potentially alleviate these concerns because control and risk are shared among participating partners. Such a digital platform, which is basically a technological manifestation of a strategic alliance, is co-owned by a consortium of platform members and collaboratively governed to orchestrate resources and co-create valued products and services (Kazan et al., 2018). The platform owners and complementors pool resources and offerings (i.e., market integration) or cooperatively create platform derivatives (Bouncken et al., 2020; Pei, 2024) to enhance consumer demand, which is beneficial to all members, especially when network effects are low and the quality difference between offerings is high (Xin & Lin, 2024). In other words, the platform owners and members cooperate to increase platform revenue through cross-selling services and offerings, cooperative advertising (Zhang et al., 2021), and subscription fees received from third parties and platform users through joint membership programmes (Yan et al., 2023). At the same time, the platform owners and members compete in capturing and sharing the revenue (Hannah & Eisenhardt, 2018; Zhang et al., 2021), and prominent members might use their superior bargaining power to capture a disproportionate share of the revenue (Bouncken et al., 2020).

Therefore, a coopetitive platform can align "the business and technology interests among platform stakeholders [. . .] and attain high levels of joint market reach" (Kazan et al., 2018, p. 213). However, if the costs of belonging to such an interorganizational platform outweigh the benefits, platform stakeholders might seek to become more monopolistic in their platform profiles, which grants control over creating and capturing business value without intervention from third parties at the platform level. As a result, they might reduce their involvement in the platform, hence reducing their effectiveness (e.g., market reach),

Table 1. Comparison of traditional and coopetitive platforms.

	Traditional digital platforms	Coopetitive digital platforms
Value architecture	The platform is owned by the platform operator, and only platform members have the right to use the platform.	The platform is co-owned by a strategic alliance or consortium of platform members who use the platform alongside other platform members and third parties.
Value network	The platform owner controls and governs the platform (i.e., monopolistic governance) and facilitates interactions and transactions between service providers and consumers within its interconnected business network.	The strategic alliance members collaboratively control and govern the platform (i.e., inclusive governance) and facilitate interactions and transactions between providers and consumers both within their pre-existing business networks and the platform's business network.
Value proposition	The platform owner and members cooperate to develop firm-specific components to co-create platform derivatives and offerings on separate layers of the platform without intervention from competitors.	The platform owners and complementors are both cooperators and competitors. They often cooperate to co-create services and offerings from complementary and pooled resources or their own services and offerings to the platform to enhance consumer demand, benefiting all members.
Value finance	The platform owner captures revenue through mechanisms such as transaction fees and subscription fees paid by platform members or through advertising, while platform members capture revenue through selling services and offerings or platform derivatives.	The platform owners and members cooperate in generating revenue through cross-selling services and offerings and cooperative advertising and subscription fees received through joint membership programmes, while they compete in capturing and sharing the revenue.

or exit the platform altogether to regain complete control over their business model (Kazan et al., 2018). In Table 1, we compare traditional and coopetitive platforms using the four dimensions of digital business models introduced in Section 2.1

3. Research design

Our main objective was to identify DPs and build a mid-range design theory (i.e., design knowledge) for developing a news platform facilitating coopetition among newspaper publishers for cross-selling online news content. To achieve this goal, we conducted a five-year [e]DSR project (Tuunanen et al., 2024) and iteratively developed the news platform through 16 design echelons. In contrast to traditional linear DSR methodologies, the [e]DSR methodology recognizes the complexity of DSR projects that often require several nonlinear DSR iterations to understand, formulate, and address real-world problems (Tuunanen et al., 2024). The [e]DSR model distinguishes five types of design echelons: 1) problem analysis, 2) objectives and requirements definition, 3) design and development, 4) demonstration, and 5) evaluation.

Figure 1 shows how our project's 16 design echelons are linked and collectively contribute to achieving a DSR project's goal of creating design knowledge through the development of an innovative IS artefact (Hevner et al., 2004; Peffers et al., 2007). Each design echelon focuses on developing and communicating an intermediate artefact, using various analytical, design, and validation activities and, as such, "makes a distinct contribution to design knowledge" (Tuunanen et al., 2024, p. 429). For instance, problem analysis echelons rely on the existing academic and professional literature and/or empirical data to generate and deliver a validated problem statement, which is the basis for further analysis and design during the objectives and requirements definition echelons to create a set of validated design objectives.

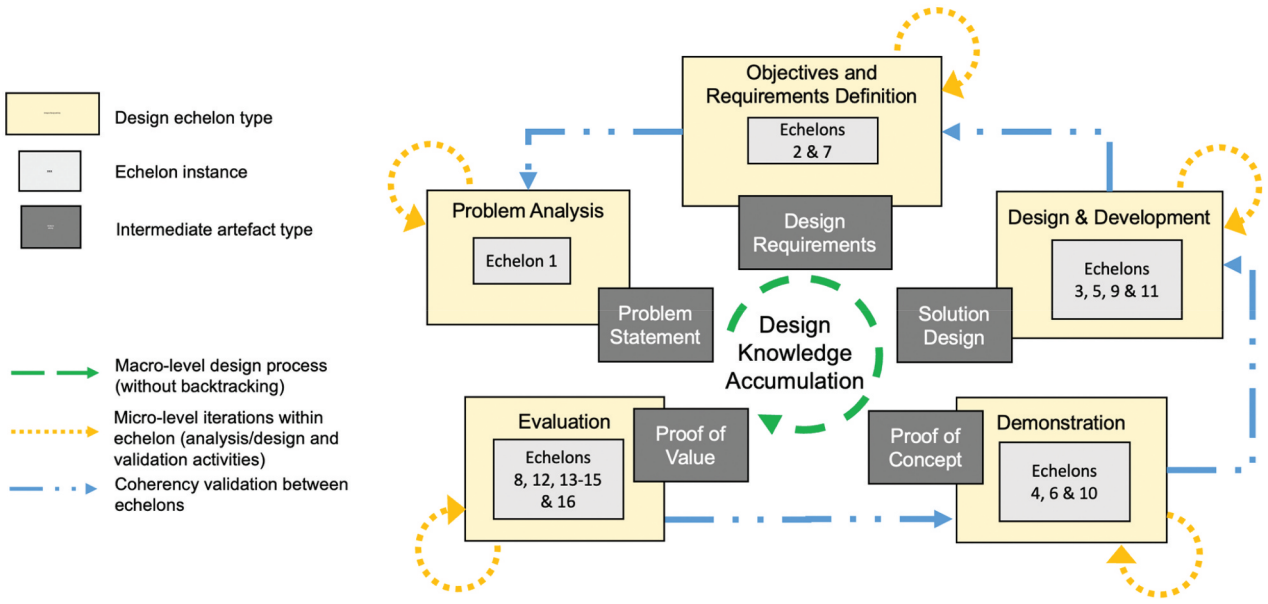


Figure 1. Project's [e]DSR meta-model depicting the design knowledge accumulation (adapted from Tuunanen et al., 2024).

4. Designing and evaluating a coopetitive news platform

This section explains how the design objectives, requirements and principles for a coopetitive news platform were developed and implemented in different artefact versions across several echelons. We applied an abductive theory building approach[3] (Sætre & Van de Ven, 2021); namely, we first deduced the design objectives and requirements based on the extant body of knowledge, which we inductively validated. This continued with deducing the initial DPs, which were then inductively validated during the following design echelons. This follows earlier examples in the DSR literature (Golovianko et al., 2023; Nguyen et al., 2021; Tuunanen & Peffers, 2018) on the abductive approach to theorization. Figure 2 provides an overview of the design echelons at the macro and micro levels. The text below elaborates on the micro-level progression of the research within each design echelon. Appendix S1, in turn, offers a detailed overview of the design echelons at the macro level of how we addressed the validation criteria in each design echelon using different validation techniques. These macro- and micro-level representations offer a complete view of how we conducted the study and how the design echelons emerged during the project. Our reporting thus follows the recommendations of Tuunanen et al. (2024).

4.1. Problem analysis

[e]DSR starts by identifying and validating a real-world problem (Tuunanen et al., 2024). In Echelon 1, we reviewed the scientific and professional literature on the newspaper industry's digital transformation and followed Tuunanen et al. (2024) to determine the degree to which our research problem had been solved and, importantly, its solvability. Our analysis indicated that digital technologies have transformed the newspaper industry and other content-based industries, such as media and music (Karimi & Walter, 2015). While the newspaper industry's digital transformation has given rise to digitized newspapers and subscription services (Winter & Alpar, 2018), it has caused major challenges and is increasingly eroding newspaper publishers' viability (Pattabhiramaiah et al., 2019). Online ad blockers, stricter privacy-enforcing platforms and regulations, competition from publicly financed news providers, readers' expectations of free content, and competition with global media houses and news platforms, such as Apple and Amazon as emerging actors in this context, have created new competition dynamics, resulting in a financially nonviable business model (see, e.g., Olsen & Solvoll, 2018).

To validate these insights, we conducted four semi-structured interviews with representatives of three Finnish newspapers and the Finnish Media

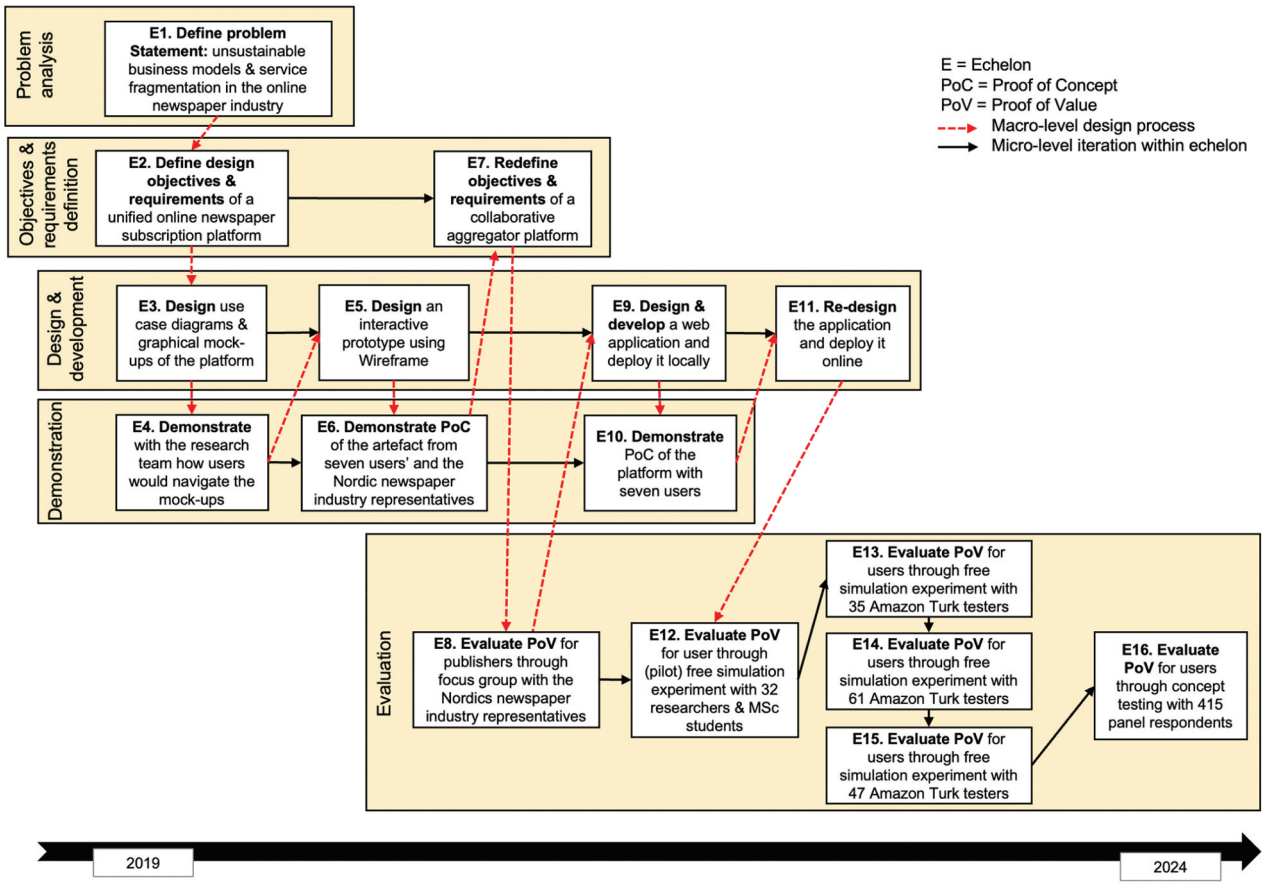


Figure 2. Overview of the FiNN+ DSR project.

Federation. These face-to-face interviews were recorded and transcribed, and thereafter, were thematically analysed. The interviewees confirmed the lack of financially viable business models as a major challenge for Finnish newspaper publishers in generating revenue from digital subscriptions to their online newspapers. The participants highlighted the challenges of monetizing online news, mainly due to the availability of free content on the Internet, as mentioned by the Chief Digital Officer (CDO) of a Finnish publisher:

> Obviously, the largest problem is the monetization part. As an industry, we have for too long given away much of the content for free, and that way, the users have been used to the fact that news is available for free. And there is a general lack of understanding in the market among the users that somebody actually needs to pay somewhere so that the news can be produced

The interviewees also highlighted the role of big digital platforms and the challenges of competing with them in developing digital solutions and capturing revenue from online content, as reflected in the following quote by the Chief Content Officer of another publisher:

> Google, MSN, Facebook, Apple are taking a major [portion] of the ads, ad revenue. And for our little nation, in a digital world, the audience is quite small if we use Finnish, because the ad euro isn't the same in digital channels as it is in print channels, so that's a problem for us. And also, the technological improvements, we should be at the same level as those big global players, and our resources are so much smaller compared to them.

Our analysis also showed that fragmentation in the online newspaper industry is another root cause of the problem. The CDO of a major newspaper echoed this concern of publishers:

> The biggest challenge is that revenues are getting smaller and smaller all the time. And the global competition is really tough, and as we heard, for example, from Apple, they are coming to the content business also. So we have a lot of competition there

Considered in the Finnish context, each newspaper has its digital subscription—a protected walled garden to cater to its reader base and derive business value. Consequently, readers initially face high adoption costs, as most users prefer to read news from several sources. The status quo, however, would require multiple subscriptions, challenging readers' willingness to pay even more. Thus, an effective design solution would include DPs for a news platform that effectively ameliorates the problem of financially nonviable business models and service fragmentation (i.e., the validated problem statement).

4.2. Objectives and requirements definition

We defined the objectives and requirements for a solution to solve the identified problem in Echelon 2. We set the following as the validation criteria (Tuunanen et al., 2024): fit to the validated problem statement and completeness according to the stakeholder requirements. We started by reviewing the literature on the newspaper industry, strategic alliance, digital platforms, and digital business models. To identify relevant studies in each of these areas, we searched different repositories, including Scopus, AIS eLibrary, and Google Scholar, with relevant keywords. For instance, to identify papers related to newspaper business models, we first searched Scopus for articles containing "newspaper" and "business model". After browsing through the most recent studies, we used a backwards and forwards search to select relevant studies that enabled us to gain a good understanding of the topic.

Drawing on the strategic alliance (Gulati et al., 2000; Hoffmann et al., 2018; Ryan-Charleton et al., 2022) and coopetition literature (Bouncken et al., 2015; Gulati et al., 2012; Nalebuff & Brandenburger, 1997), we hypothesized that to overcome financially unviable business models and service fragmentation in the Finnish online newspaper industry (i.e., a joint industry challenge) we could develop and offer a unified news platform fulfilling several objectives. First, a news platform developed by and for the Finnish online newspaper industry could enable Finnish publishers to overcome a joint industry challenge (Choi et al., 2010; Harala et al., 2023; Mathias et al., 2018) and cooperatively create platform derivatives (Bouncken et al., 2020; Pei, 2024). In addition, as platform owners and complementors pool resources and offerings (i.e., market integration) or join forces in developing and operating a unified news platform, they can overcome obstacles (e.g., lack of resources) to keeping up with technology developments and competing with big digital platforms, which is especially challenging for smaller actors, as stated by the Director of New Growth of the Finnish Media Federation:

> [the lack of] necessary resources is one of the biggest challenges especially for the smaller companies because either they don't have the skills or necessary systems that are important to make that happen or just the financial resources.

A coopetitive platform would enable cross-selling offerings, cooperative advertising, and joint membership programmes (Xin & Lin, 2024; Yan et al., 2023; Zhang et al., 2021). Furthermore, the platform would be affiliated with each partner and thus could benefit from their existing user base, considered a crucial factor for any new player in the Finnish market, as highlighted by the CDO of a small regional Finnish newspaper:

> It's always quite challenging, let's say, to build up the user base for a completely new brand. [. . .] So, probably a solution which would work better would be to tie the platform into the existing publishing platforms that are available to a certain extent.

Finally, inspired by prior studies on digital business models and pricing (Balasubramanian et al., 2015; Chen & Seshadri, 2007; Fishburn et al., 2000; Nunes, 2000; Pang & Etzion, 2012; Sundararajan, 2004; Viglia et al., 2019), enabling micropayments for purchasing single articles would allow publishers to complement their monthly subscriptions. As such, the platform would allow each publisher to increase their market reach and identify potential future subscribers, which is the ultimate goal of each publisher, as stated by the representative of a major Finnish newspaper:

> Our endgame or end goal would still be to find out the leads who like our content and try to sell them our full subscription, of course.

From the user's perspective, a unified news platform aggregating content from different Finnish online newspapers in one place would simplify the onboarding process for new subscribers. Additionally, having access to a large pool of news via one digital subscription would arguably be more attractive for readers and increase the value proposition from their point of view to pay for such a service. Finally, the news platform could enable more frequent and interactive service delivery and consumption mechanisms.

Next, we divided these design objectives (DOs) into a set of design requirements (DRs) for a coopetitive online news platform. This requirement specification was conducted through benchmarking against online news services and several rounds of logical reasoning (Tuunanen et al., 2024). We analysed several major online newspaper providers from different countries (e.g., *Spiegel.de* from Germany, *The New York Times* from the United States and *Helsingin Sanomat* from Finland). This benchmarking aimed to understand their onboarding processes (e.g., account registration and payment method) and mechanisms for accessing news articles. We also analysed various news aggregators (e.g., Google News and Apple News+) and other media subscription platforms (e.g., Netflix and Spotify) to understand their onboarding processes and pricing. Drawing on these observations and through several rounds of logical reasoning and numerous brainstorming sessions within the research team, we defined a set of preliminary DOs and DRs for designing a coopetitive online news platform. We also checked that the developed DOs and DRs met the coherency criteria of fitting the validated problem statement.

In Echelon 6, to validate the DOs and DRs, we presented them alongside our proposed solution at a workshop with seven industry experts representing the Nordic newspaper industry. The workshop discussions were recorded and transcribed. One of the authors moderated the discussion, while another author took notes during the workshop. The discussion transcript and notes were analysed thematically. The participants confirmed that the proposed solution applied to the research problem and was technically and economically feasible, as a CDO stated in the following excerpt:

> [This] is technically feasible if you put [in] enough resources [. . .]. It will obviously demand some work on the subscription management solution side.

The participants also commented on various issues related to the proposed subscription plans and collaboratively capturing and sharing revenue. For instance, the participants considered the implementation of micropayments a technically feasible option but highlighted that it should not affect the user experience of the platform, as stated by a CDO in the following quote:

> I don't see [any problems] from a technical point of view. Of course, it would be a lot of work. It shouldn't make the purchase flow harder [but] at least the same or easier.

Based on the feedback collected in the workshop, we identified barriers and common ground for developing a coopetitive online news platform and improved our DOs and DRs (see Table 2). Next, we validated the applicability of the improved DOs by conducting two semi-structured interviews with representatives of a major Finnish newspaper. The participants approved the potential of an add-on subscription and admitted that by offering only a monthly subscription, they are leaving out those customers who might be interested in reading the news but not in committing to monthly payments. However, they also mentioned that the add-on subscriptions should be offered to users carefully, as stated by the Business Development Manager of a major Finnish newspaper in the following:

> [The] risk of also offering micro-purchases is that these potential subscribers start only taking [out] these add-on subscriptions, and we will lose money in the longer term [. . .]. We would need to have [a] really good, I don't know, identifier in place [. . .] that we wouldn't always offer them but only for certain customers that we have noticed [have] some sort of behavioural pattern. [. . .]. Or then we should price them so that it's rather expensive compared to the [monthly] subscription so all the ones that are, even, likely to take the [monthly] subscription would consider that okay, this is like if I read this 4 days a month, I'm better off taking [out] the subscription

Based on these interviews, we also realized that implementing a single sign-on (SSO) would be an enabler for improving the user experience and collaboration

Table 2. Design objectives and requirements for a coopetitive online news platform.

ID	DO	DRs	Justificatory knowledge
DO1	Facilitate value creation among heterogeneous but interdependent stakeholders (e.g., content owners, journalists, freelancers, advertisers and publishers)	DR1: It should be an interorganizational news platform that orchestrates business interactions among various stakeholders DR2: It should be developed, co-owned and used by multiple publishers	• Coopetition literature (e.g., Bouncken et al., 2015; Gulati et al., 2012; Nalebuff & Brandenburger, 1997) • Strategic alliance literature (e.g., Gulati et al., 2000; Hoffmann et al., 2018; Ryan-Charleton et al., 2022) • Coopetitive digital platform literature (e.g., Bouncken et al., 2020; Kazan et al., 2018; Pei, 2024)
DO2	Simplify the onboarding process for new paying subscribers	DR3: It should provide users with different payment options DR4: It should enable users to access the service using various devices and interfaces across different online newspapers	• Design thinking • Benchmarking against online news aggregators
DO3	Increase the value proposition from the stakeholders' perspective	DR5: It should provide users with news from various publishers DR6: Subscription plans should be attractive from the readers' point of view to pay for such a subscription service DR7: Added value should be attractive from the publisher's point of view to join such a platform	• Business models for online newspapers (e.g., Chyi & Tenenboim, 2019; Sjøvaag & Owren, 2021) • Cross-selling literature (e.g., Xin & Lin, 2024; Yan et al., 2023; Zhang et al., 2021) • Product pricing literature (e.g., Balasubramanian et al., 2015; Pang & Etzion, 2012; Sundararajan, 2004; Viglia et al., 2019) • Interviews with Finnish newspaper industry representatives
DO4	Enable more frequent service delivery and consumption mechanisms	DR8: It should accommodate the continuous and dynamic flow of news articles in a timely manner DR8: It should enable users to track their balance and content consumption during their subscription period DR9: It should enable micropayments so users can buy and read articles on demand DR10: It should enable newspaper publishers to track the consumption of their content through the platform (e.g., track the attention span or amount of time a user has spent on content provided by a specific online newspaper provider) and be compensated accordingly	• Interviews with Finnish newspaper industry representatives • Workshop with Nordic newspaper industry representatives

between publishers in terms of generating and capturing revenue. Finally, the artefact's fit to the DOs was validated in four roundtable discussions with the academic community (i.e., one research seminar organized in a top European business school and three sessions at three international IS conferences).

4.3. Design and development

We started the platform prototype design and development in Echelon 3. We set the following validation criteria: applicability to instantiate the artefact, internal consistency, and justification of the design choice rationale. Through three iterations of logical reasoning and internal design thinking workshops, we developed and evaluated an architectural blueprint for the news platform, including use case diagrams and flowcharts defining different processes and sub-processes to be implemented in the artefact. First, drawing on the preliminary DOs and DRs identified in Echelon 2, we identified and analysed different use cases, describing 17 scenarios of how key stakeholders (e.g., unregistered users, registered users, and publishers) would interact with the platform for different purposes (e.g., registering, reading the news, making payments, and checking user analytics).

Next, we analysed the techniques and technologies for designing and implementing these use cases. In addition, we designed several mock-ups of the front-end interface of the platform to represent the identified use cases and workflows in the form of an intermediate artefact. This also helped us to better understand how the envisioned features should be presented to the users on different devices (e.g., mobile, tablet, or desktop computer) and how users would navigate these features.

After considering various options, we decided that a responsive web application would be the best option for addressing the operating system and device interdependence requirement, as it allows users to have a seamless experience regardless of their device, operating system, and browser. We also considered

alternatives for signing up and logging in to the system (e.g., creating a new account on the platform or using Facebook or Gmail sign-up and login services), payment methods (e.g., topping up an account with a credit card, online bank payment, cryptocurrencies, or mobile number billing), fetching and displaying the news articles (e.g., displaying full articles on the platform vs. displaying titles with a link redirecting users to the newspapers' websites). Thereafter, we designed two subscription plans (a single payment and a monthly plan) inspired by prior studies (Balasubramanian et al., 2015; Chen & Seshadri, 2007; Fishburn et al., 2000; Nunes, 2000; Pang & Etzion, 2012; Sundararajan, 2004; Viglia et al., 2019) and an SSO feature for the platform. A single payment subscription would allow users to read articles on demand and pay only for each article they accessed (e.g., €0.50 per article). A monthly subscription would charge users a monthly fee and allow them to access all news articles available on the platform without limits. The SSO would work in a similar fashion to using Gmail or Facebook accounts to log in to online services, but the SSO would be offered by the news platform provider. After signing up for the platform, users would be able to use their news platform account to access all partner publishers' online news websites.

In the next design and development echelon (Echelon 5), we developed an interactive wireframe prototype (i.e., an intermediate artefact) to test the user experience in different scenarios (Figure 3). We set the following validation criteria: applicability to instantiate the artefact, feasibility of the design, internal consistency, and justification of the design choice rationale. We evaluated the applicability, feasibility, and internal consistency of the artefact and the logic behind its components through numerous design thinking workshops and through simulating users' interactions with the designed prototype within the research team. Based on the insights gained through these activities, we validated the prototype's applicability and revised our DRs. For instance, one issue raised during the validation process related to implementing a user-friendly and fair single payment plan for users and publishers. We decided that instead of charging users for every article they accessed, they should be charged only for articles they had read. Therefore, we had to find an effective way to track a user's attention span or time spent on content provided by a specific online newspaper. Based on observations during the design and development of the validated prototype, we updated the DRs to be addressed in the following stages of the DSR project and checked that we also followed the coherency criteria (i.e., fit to the validated design objectives). In addition, following the schema[4] suggested by Gregor et al. (2020), we formulated a set of initial DPs for the proposed platform (Table 3).

In the next design and development echelon (i.e., Echelon 9) we hired two part-time software developers to support the research team in developing a web application (i.e., an instance of the artefact) simulating a news platform sourcing content from various English-speaking newspaper publishers (Figure 4). This comprised two interrelated components: a front-

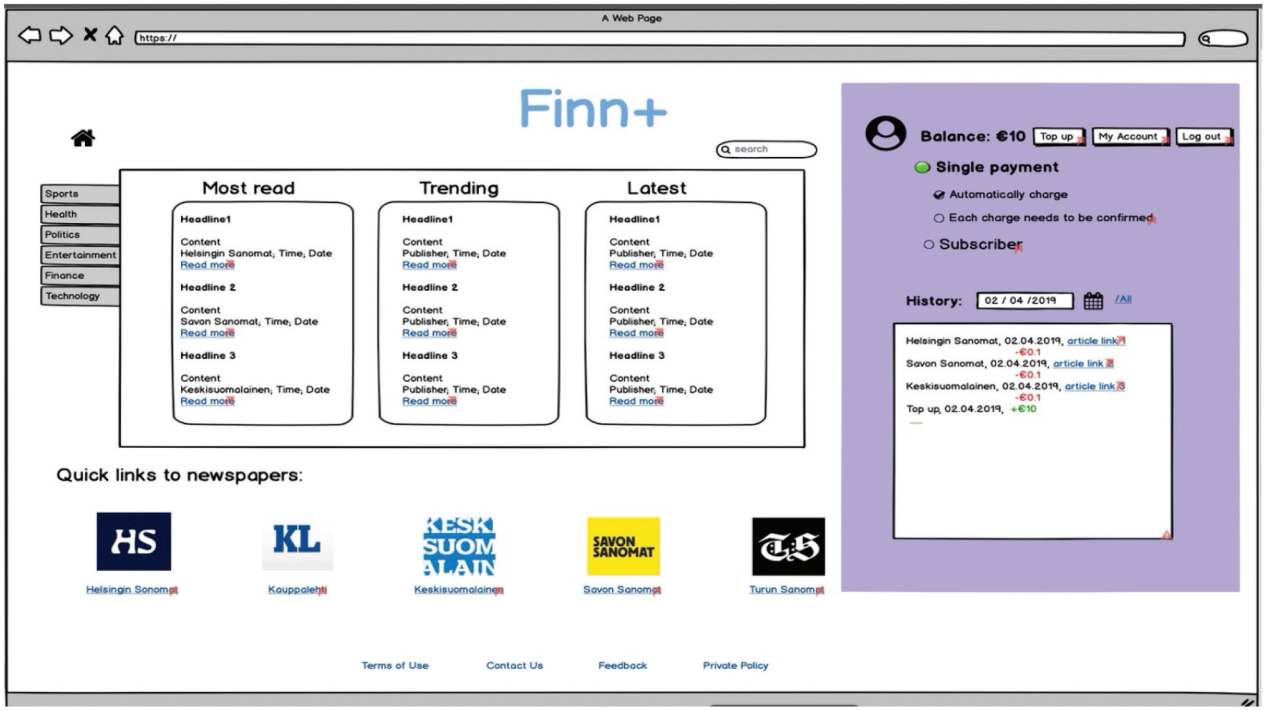


Figure 3. Snapshot of the interactive wireframe prototype.

Table 3. Initial design principles of a coopetitive online news platform.

ID	Design principle (DP)	Related design objective (DO)
DP1	Address service fragmentation for newspaper publishers by building an orchestrator hub involving online newspaper publishers to connect publishers as loosely coupled platform proxies	DO1: Facilitate value creation among heterogeneous but interdependent stakeholders (e.g., content owners, journalists, freelancers, advertisers and publishers)
DP2	Address service fragmentation for newspaper publishers by forming a coopetition alliance involving online newspaper publishers because it establishes an alliance that competes and collaborates in a hybrid business network	DO1: Facilitate value creation among heterogeneous but interdependent stakeholders (e.g., content owners, journalists, freelancers, advertisers and publishers)
DP3	Address service fragmentation for newspaper publishers by creating an add-on subscription service involving online newspaper publishers	DO3: Increase the value proposition from the stakeholders' perspective
DP4	Allow readers to access reliable news from various sources in the newspaper industry by employing a single sign-on (SSO) feature for the online news service involving online newspaper publishers to provide seamless access to news from multiple sources through one channel	DO2: Simplify the onboarding process for new paying subscribers
DP5	Provide affordable access for readers to news from various newspaper industry publishers by employing micropayments involving online newspaper publishers because it enables readers to access and pay for their favourite news articles on demand	DO4: The unified online news platform should enable more frequent news delivery and consumption mechanisms

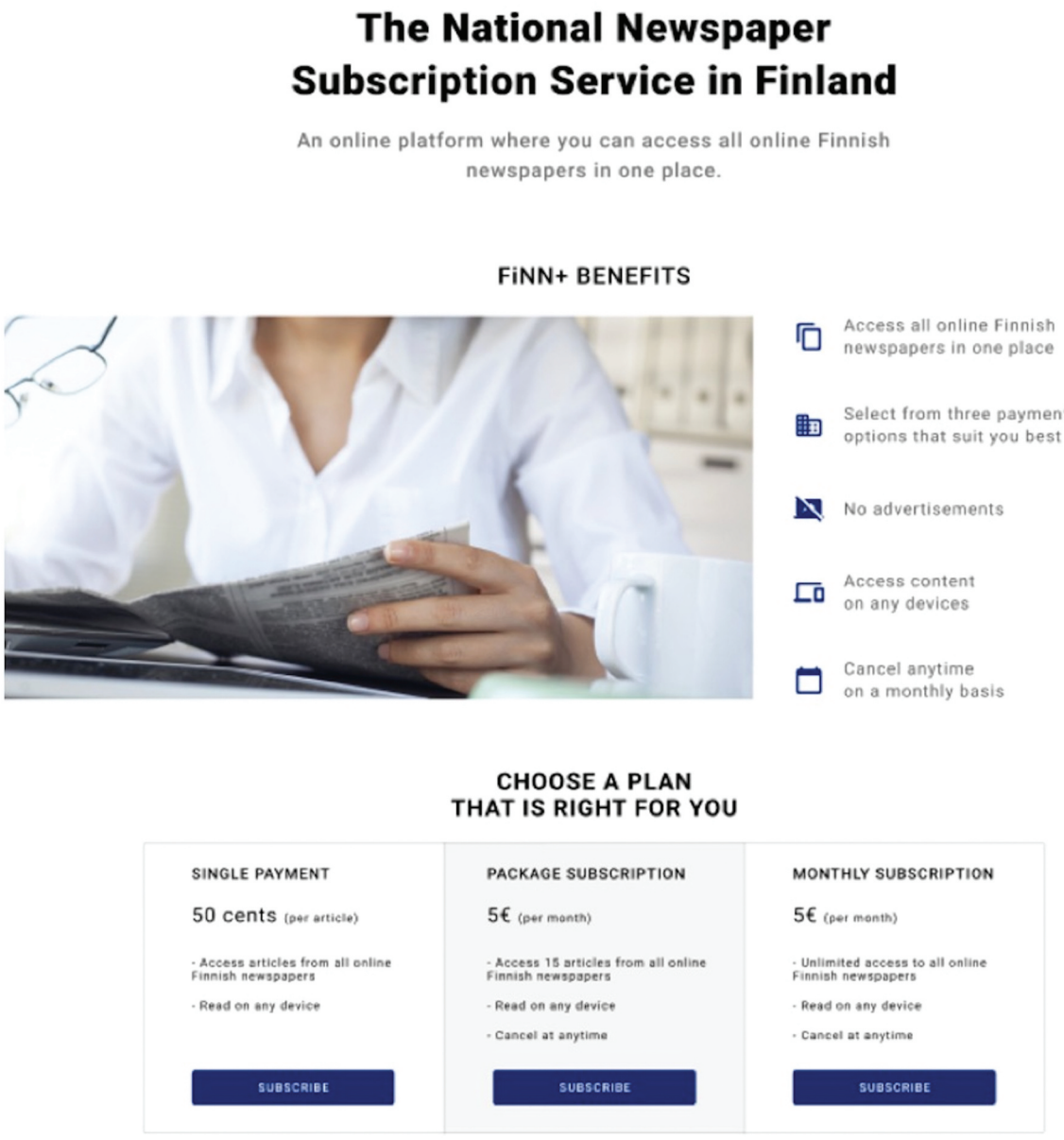


Figure 4. Main page of the prototype (see Appendix S2 for more screenshots of the platform).

end web interface that allowed users to register for the news platform, top up their account balance and access a pool of online articles by topic, popularity, or newness; and a back-end database that tracked and recorded user activities including purchases and access to different articles. As validation criteria, we set fit to the validated design of the artefact (coherency) and the functional feasibility of the artefact. Since this intermediate artefact was not publicly accessible and could not be integrated with newspapers' platforms, we manually fetched and stored in the back-end database many news articles published by Finnish and international newspapers and publishers. This was to ensure that the users testing the artefact could browse and read a wide selection of news articles, creating a natural experience like online newspapers.

We validated the coherency of the prototype and its components within our team. This was done to make sure that the features developed by the two hired software developers fitted the DOs defined and validated in the earlier echelons. In addition, we validated the functional feasibility of the artefact and its components through several rounds of testing and discussions inside the research team before demonstrating its ease of use and effectiveness in Echelon 10.

Finally, in Echelon 11, we further developed the artefact and deployed it as a publicly available web application. In addition to improving the artefact's front-end and back-end functionalities, we added a novel subscription plan (i.e., a package plan) to complement the single payment and monthly plans, reflecting existing market prices for Finnish newspaper services and subscriptions.

Furthermore, we improved the tracking mechanisms for measuring the portion of an article read by a user and charging them only if they read more than 50% of it. Regarding the payment methods, we decided to offer only credit card and online bank payments for two reasons. First, the early user testing and evaluations showed that most users selected credit card or online bank payments. Second, mobile phone billing would incur approximately 8% transaction costs for the service provider, higher than for other payment options. Additionally, we designed and implemented an information bar for the artefact containing several functionalities. The bar provided readers with data about their current payment plan, the number of articles left to read, at what point they would be charged for the current article they were reading and options for sharing the article with others or adding it to their favourites list (Figure 5).

Moreover, we integrated an RSS feed into the artefact to fetch and show the latest news articles from the selected publishers to offer a more realistic user experience to our test users. Finally, since the updated artefact version was intended to be deployed and hosted on a web server, making it publicly accessible to potential users, we created terms of use and privacy policy documents based on local regulations. The applicability and internal consistency of the artefact was validated through several rounds of user testing within the research team before evaluating the artefact with potential users in Echelons 12–16.

4.4. Demonstration

We conducted three demonstration echelons. First, we demonstrated the graphical mock-ups' fit with the DOs and DRs in Echelon 4. This was done through iterative testing of the graphical mock-ups within the research team to ensure that the envisioned features enabled users to navigate and complete the identified workflows and use cases.

In Echelon 6, we demonstrated the functional and economic feasibility of the artefact and validated its preliminary proof of concept (PoC) from users' and publishers' perspectives. We first asked seven potential users (i.e., researchers from our network) to interact with the wireframe prototype and perform tasks such as creating and topping up user accounts and accessing paywalled content and wallet functionalities. This user testing aimed to ensure that the design and its

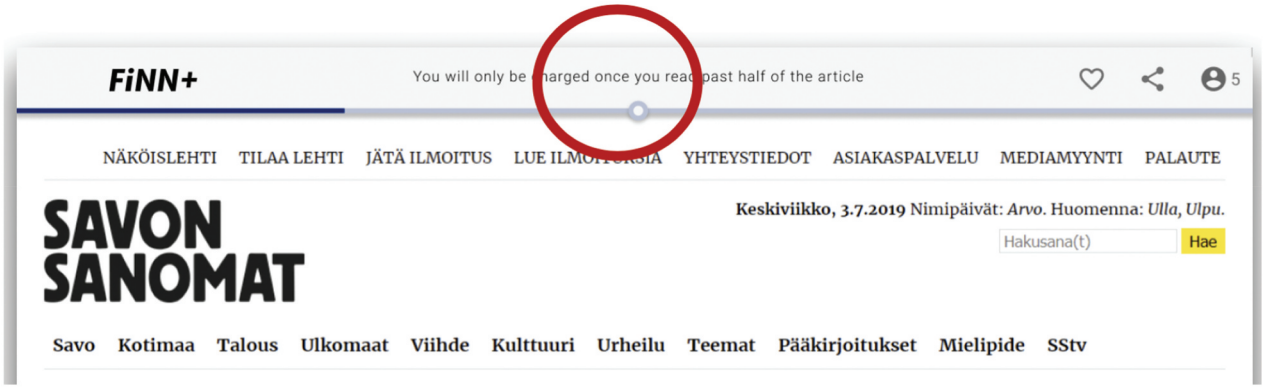


Figure 5. Snapshot of the information bar added to the web application. The red circle highlights the 50% threshold when users have read half of an article and indicates that they will be charged if they read more.

underlying features and mechanisms were logical and easy to understand for users, while feasible for addressing the identified and emergent DOs and DRs.

Next, the artefact's economic feasibility was demonstrated and discussed as part of a workshop with a group of representatives from leading Nordic newspaper publishers. The participants conveyed the notion that the proposed solution should be complementary to their own services without resulting in cannibalization effects. For instance, a small local publisher's representative believed that the proposed platform would provide additional exposure to their regional content. In contrast, a major national publisher's representatives mentioned that the platform would enable them to generate extra revenue on top of their existing subscriber base as an add-on service. Another participant emphasized that "*It has to give us revenues [and] not eat our customer base*". To that end, we proposed micro-purchases of single articles on demand as an extra revenue stream, which the participants considered a viable option:

> If someone goes to publisher A's property, reads and pays for it, publisher A gets the money [minus] a service fee; that is very straightforward.

However, they mentioned that this should not degrade the user experience and make the purchase flow harder. Another publisher representative conveyed his concerns: "*If you sell single [articles] or day passes versus [monthly] subscriptions, the maths is completely different; [. . .], we measure every subscription by net present value, and it is so much higher [compared] to individual articles. My Excel [spreadsheet] doesn't give me [the] green light, at least yet*". Nevertheless, the workshop provided us with preliminary insights that the proposed solution could function as an add-on subscription service that builds on existing publisher-controlled newspaper subscriptions.

In Echelon 10, we demonstrated the ease of use and effectiveness of the locally deployed web application designed in Echelon 9. First, we asked seven researchers from our network to assess the news platform's functional feasibility and ease of use from a potential user's perspective. Then, we conducted a workshop with these users and discussed their experience with the artefact. Their feedback was generally positive (e.g., *"Great concept"* and *"I like the favourites option, as it allows you to save interesting or unread articles for later reading"*) and constructive, as they spotted a few issues related to the user interface and the artefact's functionality. For instance, one user spotted an inconsistency between the visual and textual elements when accessing the artefact from a desktop computer or smartphone. Other users reported bugs related to the registration process (e.g., *"For some reason, the registration page only works if I first input my password and then the rest of the information"*.) and navigating between pages (e.g., *"The browser's back button takes you back to the general front page instead of, for example, the Technology page if you had that open"*).

The users also provided suggestions for improving the usability and user experience. For instance, one user recommended improving the navigation between articles by integrating a back button into the article reading view, making navigating easier. Others suggested adding more details to some menus to make the interaction easier. For instance, one user suggested showing the price of each subscription in the drop-down menu used to choose a subscription plan during registration. Another user suggested highlighting the source of each article listed in the news feed to help users evaluate the trustworthiness of an article before opening it: "*In these times of fake news, at the least, I evaluate what I read based on which newspaper/agency/reporter has written it, and often I don't even open an interesting sounding article if it is published in media I have little trust in*". This feedback enabled us to validate the coherency criteria (i.e., fit to the validated design of the artefact) and PoC and improve the artefact based on the collected feedback and suggestions.

4.5. Evaluation

To validate our artefact's proof of value (PoV) from publishers' and users' perspectives, we conducted three evaluation phases with the following validation criteria: fit to the validated instance of the artefact (coherency) and the artefact's efficacy, projectability, and utility. First, we organized a workshop with the Nordic newspaper industry representatives to validate DP1 and DP2 (i.e., Echelon 8). Second, we conducted a set of free simulation experiments (i.e., Echelons 12–15) with potential users of the news platform to evaluate DP3 and DP5. Finally, we conducted a concept testing survey (i.e., Echelon 16) to further validate the findings of the free simulation experiment and evaluate DP4.

4.5.1. Workshop with online newspaper publishers

To evaluate the PoV from online newspaper publishers' perspectives and validate our DPs, we relied on the feedback provided by the representatives of leading Nordic newspaper publishers and the Finnish Media Industry Association (Echelon 8). As discussed earlier, in Echelons 6 and 7, we organized a workshop as part of the media industry association's board meeting. In the workshop, we first demonstrated our proposed solution and then collected feedback related to the technical and business aspects pertinent to micropayments and coopetitive subscriptions. The participants' feedback and insights enabled us to validate DP1 and DP2 and gather ideas to further improve other DPs. The participants recognized the need for coopetitive

platforms to address service fragmentation for newspaper publishers (DP1) and the need to form a coopetition alliance of online newspaper publishers (DP2).

During the workshop, the participants told us that the Finnish Media Industry Association was working on a collaborative project to develop an SSO capability for their member newspapers so that their subscribers could sign in to another online newspaper with the same account. At the time of the workshop, the project had gained funding of several million euros, and we were informed that the project would commence within the next few months. We identified two ideas for improving: 1) the mechanism for tracking and billing users' article "consumption" and 2) an extra viable coopetitive subscription plan that would be attractive for various publishers (e.g., small and large publishers with local and national reach). Regarding the former, we proposed that instead of tracking users' reading time, we would track the percentage of the article read by users and charge accordingly (e.g., reading the first 50% of the article would be free, and users would be charged only after reading more than 50% of an article).

Regarding the digital subscription model, one participant brought up the "dividing the cake problem", implying that the proposed subscription plans (i.e., single or monthly payments) would not be attractive for more prominent publishers in Finland, since they already had a large user base with monthly subscriptions. All the major newspaper participants rejected the fixed flat-rate subscription; they would not be interested in sharing their content with smaller newspaper readers for free. Two ideas for addressing this issue were proposed. First, the prices for reading single articles could differ, for instance, based on the topic or the publisher, instead of having a uniform pricing scheme. This feature would enable each publisher to influence the single payment option based on their business strategies (e.g., to sell more articles at lower prices or fewer articles at higher prices) and the amount of effort spent on preparing an article (e.g., a finance article might require considerably more research than a news article about a local event). Second, a third subscription option would allow subscribers to access a limited predefined number of other newspapers' articles by paying an extra monthly fee (e.g., €5). The participants unanimously agreed that this option (i.e., a package plan) could work in the Finnish online newspaper market, as it would allow each publisher to maintain customer relationships with their base subscription plan while opening up incremental new business opportunities by attracting other publishers' readers.

4.5.2. Free simulation experiment

To evaluate the PoV of the web application developed and deployed online from the user's perspective and validate DP3 and DP5, we combined formative and summative evaluations (Tuunanen et al., 2024). We conducted a formative evaluation to identify areas of the artefact that should be further developed and a summative evaluation to assess the overall user experience with the artefact and the value proposition of the proposed news platform. We conducted a set of free simulation experiments in four evaluation echelons (i.e., Echelons 12–15). Each experiment comprised a series of tasks and steps a user had to complete to be able to access and read the news through the platform.[5]

Free simulation experiments expose participants to real-world-like events and environments within a specified timeframe (Fromkin & Streufert, 1976; Streufert et al., 1968). While completing the experiment within the boundaries specified by researchers, the participants have the freedom to complete those tasks in an unrestricted manner. Therefore, realistic events during the experiment occur as the participants interact with and explore the prototype and behave within the simulation environment on their own, which, in turn, provides them with a better understanding of the experimental artefact (Gefen et al., 2003; Söllner et al., 2016).

Following previous studies' suggestions (Fromkin & Streufert, 1976; Gefen et al., 2003; Söllner et al., 2016; Streufert et al., 1968), we designed the experiment setting so that each participant could interact with the prototype on their own and independently from other participants. However, to complete the experiment, the participants had to follow a set of steps within the specified time and boundaries. First, each participant had to visit the platform and sign up for the service. Next, they had to choose a subscription plan and their preferred payment method. In the next step, the participant could start navigating different areas of the prototype and explore and read news articles. After completing these tasks, the participants had to fill in a short questionnaire, including demographic questions and questions about their experiences of using the artefact.

To guide these experiments, we developed a protocol that included descriptions of the proposed news platform, the tasks to be conducted and the compensation for completing the tasks. We asked the participants to sign up for the service after familiarizing themselves with the value proposition of the platform and to choose one of the proposed subscription plans that would be sufficient to fulfil the task, while at the same time considering their budget and willingness to pay for this type of plan. As part of this, we designed and allocated an imaginary budget of $7.50 to each tester for purchasing and reading news articles. We carefully designed and reviewed this imaginary budget and the amount of compensation based on the three subscription plans (i.e., single, package, and

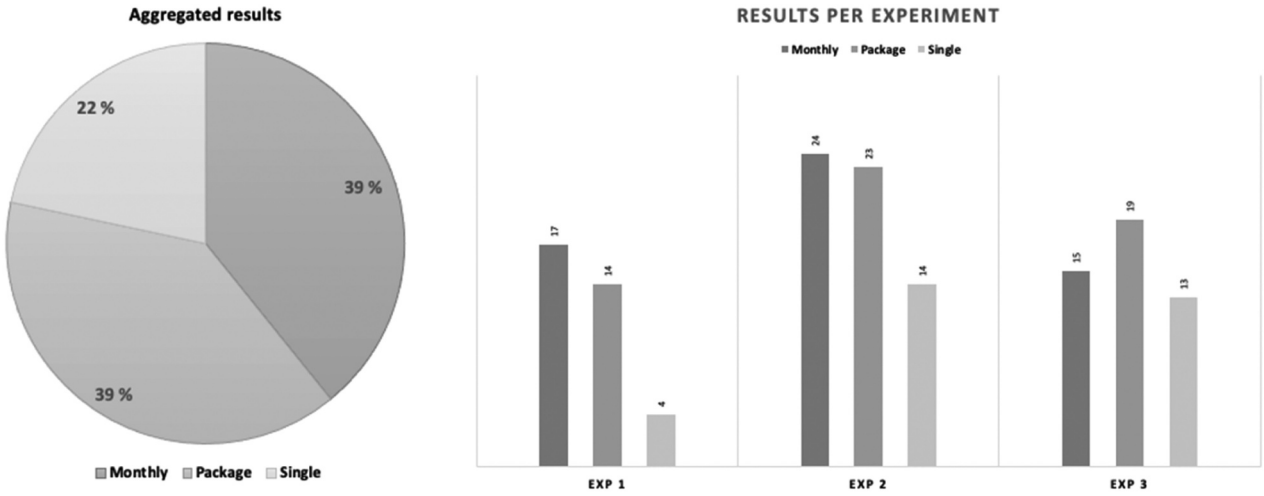


Figure 6. Overview of the results of the experiments.

monthly plans) so that the maximum compensation for the participants was always $5, no matter which plan was chosen or how many articles were read[6]; namely, we offered the participants a fee for summarizing 5 to 10 articles ($0.75 per article). In addition, we incentivized the participants by offering a small fee for just reading a maximum of five additional news articles ($0.15 per article). This was done to simulate the value that we, as news readers, gain from reading news articles without making a micropayment every time. We offered this for both the package and monthly subscriptions but not for the single payment subscription. From the results, we can see that this manipulation worked well. The participants were informed that the subscription price would be deducted from their rewards before payment. Our goal was to deter the participants from gaming the experiment for a higher economic benefit and to determine their value perceptions and individual estimations of their expected use. Ultimately, our focus was on evaluating the news platform prototype's usage and the performance of the different payment plans.

After preparing the protocol, we conducted the experiments in two stages with a total of 175 participants who, demographically speaking, were among the target users of the news platform. In the first stage (Echelon 12), we conducted a pilot test with 32 researchers and master's students from our network who were not offered any incentives or rewards. After this stage, we improved the protocol by adjusting the description of the work task for Amazon Mechanical Turk (MTurk) workers; namely, the number of articles the workers needed to submit to be compensated in full. We then recruited 143 testers (aged 25–35 years) through MTurk. For the first experiment (Echelon 13), we recruited 35 testers; for the second experiment (Echelon 14), we recruited 61 testers; and for the third experiment (Echelon 15), we recruited 47 testers. After each experiment, we checked the results and made some further minor adjustments to the experiment protocol, while the pricing model for the subscriptions remained the same in all three experiments. The difference between the first and second experiments was how the compensation to the MTurk workers was communicated on the experiment's web page. In the first experiment, we noticed that some workers needed further instructions for this. During the second experiment, we noticed a few software bug ithat hindered the use of the online news platform, namely how the news content was uploaded to the platform.

In addition to collecting quantitative data and statistics through the prototype (e.g., information about the users' choices and interactions on the platform), we collected qualitative data from the participants through a survey for two reasons. First, to ensure that the testers completed the tasks according to the instructions, we asked each tester to write a summary of 2–3 sentences for a minimum of 5 and up to 10 news articles they had read. Second, we asked them to provide feedback on their experiences with the prototype and their reasons for choosing a specific subscription plan.

The results show (see Figure 6 left) that, from the total sample of testers, 56 (39%) selected the monthly subscription, 56 (39%) chose the package subscription and 31 (22%) selected the single article option, meaning the add-on subscription (i.e., single and package plans) was selected by the majority of the participants (61%). These patterns are reflected in the results of each experiment shown on the right of Figure 6. When looking more closely at the results of echelons 13–15, we notice that after the experiment setup was iteratively finalized in echelon 15 the majority of the participants chose the package plan (40.4%). These results validate our proposal that offering an add-on subscription through the single and package plans is an attractive option for generating extra revenue for publishers (i.e., DP3), as it enables them to attract new

Table 4. Qualitative results from the online experiments.

Selected subscription	Sample answers provided by testers for choosing the subscription
Monthly	• I chose the monthly subscription because it gives me the most articles to read, and I don't have to worry about running out. • I read articles daily, and I would save more money with the monthly plan instead of paying for each article or only 15 articles. I tend to read around three to five articles daily, so it would make sense to use this payment plan. • I think the value of being able to read more than 15 articles a month is worth the price increase of about 3 dollars.
Package	• I picked the package because 15 articles a month is sufficient for me from one news site since I like to get my news from many different sources. • I chose the plan that I did because it seemed that this amount would allow me to specifically focus on news articles that I would solely be interested in. • I chose this payment plan because it is what I needed. I didn't want to choose a plan that I had to overpay for something I would not use as much as I would have intended. It saves me money.
Single article	• I chose the smallest plan because I often get my news in either a visual or audio way; I don't read all that much, so having a small, cheap plan works best for me. • I chose the single plan because I want to test out the platform and see if it's right for me before deciding further. • I wanted to just click on the articles I really wanted to read, so if I paid per article, I would make sure that I chose only the most interesting-looking articles.

users who would like to read a limited number of articles from various newspapers at an affordable price. This result also impacted DP3, which was revised accordingly (see Table 7, with the added text in bold).

Looking at the qualitative data and the reasons the testers provided for choosing each package (Table 4), each package had unique features that attracted a specific group of users. Most users who selected the monthly subscription seemed to read news frequently. They highlighted the unlimited nature of the plan as the main reason for choosing it. The users who selected the package plan highlighted its value-for-money nature. This group of users seemed to consume news articles from different sources, and therefore, found the limited number of articles in the package plan sufficient for their needs. Finally, the users who selected the single article plan highlighted its flexible nature, allowing them to pay only for those articles they were interested in. In other words, this plan enabled users to read news on demand to avoid the costs of committing to a specific subscription. These results validated our original DP5.

4.5.3. Concept testing market survey

To further validate the generalizability and projectability of the findings of the experiments and to evaluate DP4, we decided to conduct a concept testing survey in Echelon 16 to capture the subscription preferences of potential platform users. Concept testing is a market research method for capturing potential users' reactions to a product concept before its launch and exploring their willingness to buy it (Ozer, 1999; Peng & Finn, 2008; Reid & Crawford, 2022). To conduct the test, we designed a survey which started with a brief overview of the news platform and its value proposition.[7] Next, the respondents were asked to indicate their purchase intent by rating the attractiveness of each of the three subscription plans on a 5-point Likert scale (1 = most likely not choose to 5 = most likely choose). In addition to standard concept testing questions, we asked the respondents to rank the subscription plans based on their attractiveness if they would have joined the platform (rank 1 = most attractive option and rank 3 = least attractive option). The respondents were also asked to rate on a 5-point Likert scale the importance of the SSO feature if they would have joined the platform (1 = not important at all to 5 = very important). Finally, the respondents answered two questions about their news consumption frequency on a 5-point Likert scale (1 = never to 5 = daily) and their main sources of news.

We recruited participants through the Conjointly platform from a predefined panel of respondents interested in news. We launched the survey in June 2024 and received 541 responses, which, after the quality check, resulted in 415 complete responses (298 responses from the UK and 117 from Germany). Most of the respondents read news frequently (69% daily). In addition, over 55% of the respondents obtained news from online news websites, and over 21% read print newspapers. Therefore, the sample can be considered representative of potential users of the platform.

In terms of purchase intent, 53% of the respondents reported that they were likely or most likely to choose the monthly plan if they joined the news platform. Both the single article and package plans were reported to be likely or most likely chosen by over 49% of the participants. This is somewhat like the results of the free simulation reported in Section 4.5.2 Moreover, 64% of the respondents considered both the package and single article plans as new and different from the subscription options already available in the news market. This confirms the novelty of these plans for supplementing

Table 5. Respondents' purchase intent per subscription plan.

	Single article plan	Package plan	Monthly plan
Whole sample (n = 415)	49.6%	49.4%	52.8%
UK sample (n = 298)	49.0%	51.7%	53.4%
Germany sample (n = 117)	51.3%	43.6%	51.3%

more common monthly subscriptions. Table 5 summarizes these results.

In terms of attractiveness, the monthly subscription was ranked the most attractive option by just over 45% of the respondents. This was not surprising, since the participants did not need to consider how to use an imaginary budget most efficiently, as in the free simulation. While the single article plan was ranked the most attractive option by just over 31% of the respondents, the package plan was ranked the most attractive option by a slightly lower proportion of participants (i.e., just over 23%). However, while the package plan was the second most attractive option for just over 48% of the respondents, the single article plan was the least attractive option for just over 46% of the respondents. Figure 7 summarizes the results.

These results support our argument that an add-on subscription, such as a package plan or single plan, could complement the monthly plan. This is in line with the results of the free simulation experiment, especially since we had earlier learned that the major newspaper publishers clearly rejected offering a fixed-price monthly plan for the platform, as they saw this cannibalizing their current revenue.

Lastly, regarding the value of the SSO feature, 61% of the respondents reported that the feature would be important or very important for them, while 15% of the respondents reported that the SSO would not be important for them. The results also enabled us to check that we followed the coherency criteria (i.e., fit to the validated instance of the artefact) with the design echelons. Looking at the qualitative data and the reasons the respondents provided for choosing each package or the importance of the SSO (Table 6), the results further support the findings from the free simulation experiments.

The most important factors mentioned by the participants for preferring the monthly subscription plan were value for money and the convenience of knowing the fixed cost upfront while having unlimited access to articles. They especially appreciated the flexibility to read as many articles as they wanted without additional charges. The most important factors that the participants mentioned for preferring the package subscription plan were its cost-effectiveness and the freedom to read a variety of articles. The respondents seemed to appreciate the convenience of a single upfront payment that provides access to multiple articles from various publishers. The most important factor mentioned by the participants for choosing the single article plan was the flexibility of only paying for the articles they were interested in reading without committing to a monthly subscription. Many respondents seemed to find value in the ability to select and

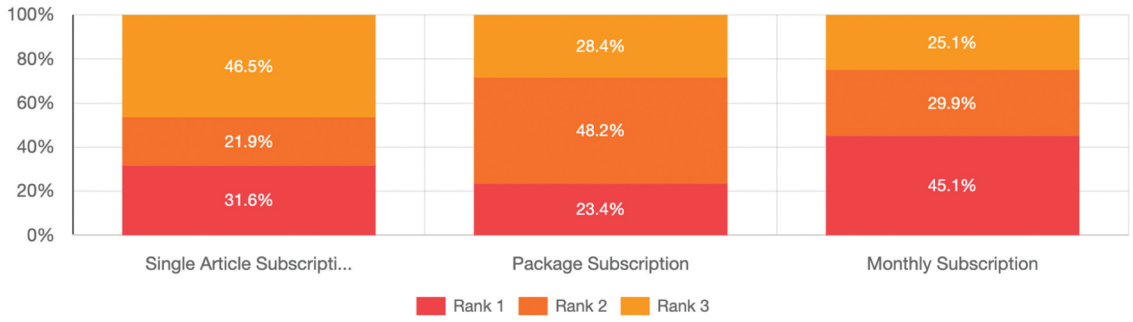


Figure 7. Overview of the concept testing survey results.

Table 6. Qualitative results from the concept testing.

Selected subscription	Sample answers provided by the respondents
Monthly	• Because it's unlimited, you know exactly how much it's going to cost. • You know in advance how much you will pay per month, and you can read an unlimited number of news articles. • I would likely read more articles per day than the more limited offer provides, and the individual article offer would work out quite expensive.
Package	• I can choose the articles I'm interested in and pay only for those. • It's more control of how many articles you can read in the timeframe given. • It is a cheaper alternative to the monthly subscription, and I don't think I'd read more than 15 articles a month. • It provides a good range of articles to choose from for a fair price.
Single article	• Only paying for the articles that interest me with no commitment. • I get to pay only for the articles I read. • I think it would be a cheap option, and I would just be paying for articles I really want to know more about. • You are only paying if you actually want to read an article.
SSO feature	• It would make signing in easier. • It makes the process seamless and easy. • The convenience, the user experience and security. • I don't have to worry about remembering passwords. • It's not really important. Would be good to have as a security feature though. • It's not really important. It's a feature I could live without.

Table 7. Final design principles for designing a coopetitive news platform and their justificatory knowledge.

ID	Design principle (DP)	Justificatory Knowledge
DP1	Address service fragmentation for newspaper publishers by building an orchestrator hub involving online newspaper publishers to connect publishers as loosely coupled platform proxies	• Coopetition literature (e.g., Bouncken et al., 2015; Gulati et al., 2012Nalebuff & Brandenburger, 1997) • Strategic alliance literature (e.g., Gulati et al., 2000; Hoffmann et al., 2018; Ryan-Charleton et al., 2022) • Coopetitive digital platform literature (e.g., Bouncken et al., 2020; Kazan et al., 2018; Pei, 2024) • Empirical research in Design Echelons 1–10, including prototype demonstration, focus group, and workshops and industry interviews
DP2	Address service fragmentation for newspaper publishers by forming a coopetition alliance involving online newspaper publishers because it establishes an alliance that competes and collaborates in a hybrid business network	
DP3	Address service fragmentation for newspaper publishers by creating an add-on subscription plan involving a predefined package of content from various online newspaper publishers because it bundles limited add-on subscriptions with unlimited proprietary ones (i.e., base subscriptions)*	• Business models for online newspapers literature (e.g., Chyi & Tenenboim, 2019; Sjøvaag & Owren, 2021) • Cross-selling literature (e.g., Xin & Lin, 2024; Yan et al., 2023; Zhang et al., 2021) • Product pricing literature (e.g., Balasubramanian et al., 2015; Chen & Seshadri, 2007; Pang & Etzion, 2012; Sundararajan, 2004; Viglia et al., 2019) • Empirical research in Design Echelons 7–16, including prototype demonstration, focus group and workshops, industry interviews, free simulation experiment, and concept testing survey
DP4	Allow readers access to reliable news from various publishers in the newspaper industry by employing a single sign-on (SSO) capability for the online news service involving online newspaper publishers to provide effortless access to aggregated news from multiple sources in one place	• Design thinking and benchmarking against online news aggregators • Empirical research in Design Echelons 1–8 and 16, including prototype demonstration, focus group and workshops, industry interviews, and concept testing survey
DP5	Provide affordable access for readers to news from various newspaper industry publishers by employing micropayments involving online newspaper publishers because it enables readers to access and pay for their favourite news articles on demand	• Empirical research in Design Echelons 1–15, including prototype demonstration, focus group and workshops, industry interviews, and free simulation experiment

*The bold text indicates the revisions made to the DP.

pay for individual articles, making it a more personalized and economical option. Lastly, the most important factors mentioned by the participants regarding the importance of the SSO feature were its convenience and ease of use, allowing users to quickly access multiple accounts with one log-in. The participants suggested that this improves the user experience and security by reducing the need to remember multiple passwords. Even many of those respondents who indicated that the feature was not particularly important to them, considered it a "good to have" feature.

In the next stage, we conducted multigroup analyses to examine the potential differences between various target customer groups. In terms of geographical region, a higher proportion of respondents from the UK (51%) compared to Germany (43%) reported that they would be likely or most likely to choose the package plan. Second, a considerably higher proportion of respondents from Germany (40.2%) ranked the single article plan as their first option compared to the UK respondents (28.2%). Similarly, a considerably higher proportion of respondents from the UK (26.5%) ranked the package plan as their first option compared to those from Germany (15.4%). In terms of news consumption frequency, the participants who read news regularly (but not daily) tended to choose the package plan, while the ones who read news daily tended to choose the monthly plan, and those who read news rarely tended to choose the single article plan. This is in line with the qualitative comments provided by the respondents and further supports the findings from the free simulation experiments. The results of a t-test showed that the respondents who were *most likely* to choose the package plan, on average, consumed more news than those who were likely to choose the package plan ($p < 0.05$). In other words, the respondents who consumed more news were more likely to choose the package plan (within this user group). Lastly, comparing the respondents from different age groups and earnings, the respondents between 31 and 40 years old and the ones with an annual income between 19,500€ and 65,000€ had the highest tendency to choose the package plan.[8]

To conclude the DSR project, we prepared a report to communicate the results and discuss avenues for future implementation of the artefact with Finnish newspaper industry representatives. In addition, we provided the practitioners with access to the artefact's source code for future use and later uploaded the source code to GitHub. In the next section, we discuss our final DPs and the contributions of the study.

5. Discussion

In this five-year study, we aimed to propose DPs for a news platform that supports coopetitive cross-selling of online news content. Through our close collaboration with the Finnish Media Industry Association and Finnish newspaper publishers, we have generated generalizable design knowledge for developing a coopetitive news platform to address the lack of a viable business model for monetizing news content and generating steady revenue (i.e., a joint industry-level problem). Next, we present how we abductively build DPs through this industry collaboration that contribute to a mid-range design theory for designing

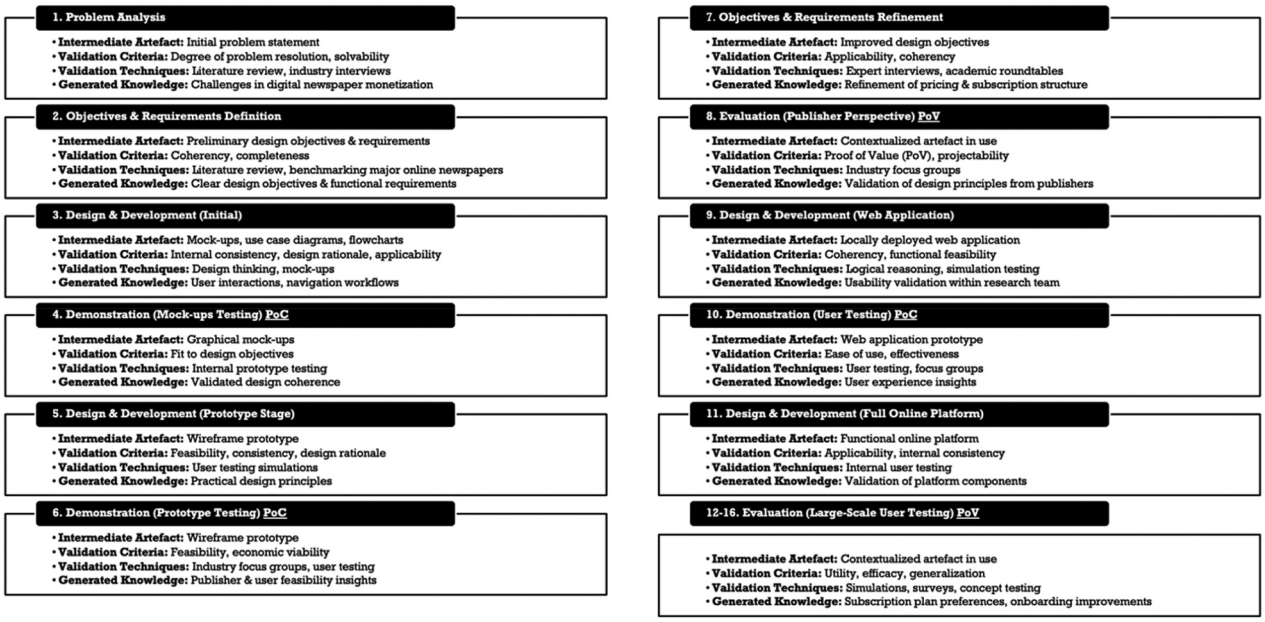


Figure 8. Abductive design theory building in the FiNN+ project.[9]

coopetitive news platforms. Thereafter, we state our contributions to the coopetition and online newspaper literature, and DSR practice.

5.1. Abductively building a mid-range design theory

Our abductive design theory building process is summarized in Figure 8. Abductive theory building is "a cyclical process of identifying and confirming anomalies [i.e., problem] and generating and evaluating hunches [i.e., explanations] to make sense of complex phenomena" (Sætre & Van de Ven, 2021, p. 686). According to Sætre and Van de Ven (2021), this iterative sensemaking process has four steps (i.e., observe anomaly, confirm anomaly, develop hunches, and evaluate hunches), which were conducted in our project as follows.

We began by identifying the problem statement in Echelon 1 through a literature review and internal discussions, observing that newspaper publishers struggle to monetize online content (i.e., step 1 of the abductive theory building). The problem was then confirmed empirically via interviews with representatives of the Finnish online newspapers industry (i.e., step 2 of the abductive theory building). Having validated the problem, we proceeded with developing ideas for identifying potential solutions to address the identified problem (i.e., step 3 of the abductive theory building) by moving back and forth between theory (e.g., strategic alliance and coopetition literature) and practice (i.e., design and development activities). In terms of ^{e}DSR, this was done through defining and refining DOs and DRs (cf. Table 2) for potential solution designs in Echelons 2 and 7 followed by developing several instances of the coopetitive news aggregator solution in Echelons 3, 5, 9 and 11 to generate initial DPs (cf. Table 3).

Finally, to evaluate our design knowledge (i.e., step 4 of the abductive theory building), we iteratively validated the proposed design artefacts. In ^{e}DSR terms, we first demonstrated the PoC of the proposed solutions and DPs in Echelons 4, 6 and 10. Then, we evaluated PoV of the proposed solution in echelon 8 and 12–16. The validated final DPs summarized in Table 7.

Our mid-range design theory consists of new design knowledge for designing coopetitive digital platforms accumulated during this project. Our formulation is more attuned to the nature of mid-range theory defined by Gregor and Hevner (2013) as new design knowledge about an embedded phenomenon, accumulated through situated instantiations of the proposed solution, developing more abstract design principles, and providing a better understanding of the problem and solution spaces. This is in line with how Merton (1968) characterized a mid-range theory as new design knowledge derived deductively from the reference theories and inductively from data collection and analysis explaining a limited set of phenomena (see Figure 8). Accordingly, our mid-range design theory description follows DiMaggio's (DiMaggio, 1995) "theory as a narrative" conceptualization that takes the position of emphasizing how a theory can offer a plausibility of the socially constructed narrative in a situated condition. The mid-range design theory is summarized as a set of theoretically grounded and empirically evaluated prescriptive DPs (see Table 7), which are considered "as the most distinctive part of a design theory" (Gregor et al., 2020, 1622).

Although our mid-range theory is specifically formulated for designing a coopetitive news platform, we believe its utility can extend beyond the newspaper industry for designing other instances of coopetitive digital platforms which their main value proposition is offering creative content to consumers. Our project was motivated by a real-life problem: the lack of financially viable business models for monetizing online newspapers. Upon further analysis we discovered that service fragmentation in the online newspaper industry alongside competing with big digital platforms or free news websites were the main root cause of the problem. Previous research shows that other content-based industries, including film, music, and game industries, face similar challenges caused by digitalization (Karimi & Walter, 2015). For instance, digitalization has reduced the artists' and music producers' revenues due to piracy or unauthorized file sharing (Tilson, 2013; Waldfogel, 2012) and consumers' reluctance to pay for music when they have access to free samples and trials (Jensen, 2024). In addition, monopolistic digital platforms have dominated the music industry (Handke, 2020; Tofalvy, 2023) and exacerbated the problem as they challenge the traditional business models of music companies (Eriksson, 2019; Poell, 2020) and led to perceived inequities in the distribution of streaming revenues by artists (Jensen, 2024; Tofalvy, 2023). Drawing on lessons learned from our study; we propose that a coopetitive digital platform could mitigate these challenges.

As reflected in DP1, we propose that a coopetitive digital platform can serve as an orchestrator hub for connecting content (e.g., news, music, videos, games) producers and publishers as loosely coupled platform proxies. As reflected in DP2, forming a coopetition alliance enables various content producers and providers to simultaneously cooperate for common benefits and compete for private benefits (Gulati et al., 2012; Nalebuff & Brandenburger, 1997; Stadtler & Van Wassenhove, 2016; Yadav et al., 2022). As such, the coopetitive platform not only enables publishers to extend their market reach (Kazan et al., 2018) but also to co-create value (Bouncken et al., 2020; Kazan et al., 2018; Pei, 2024), for instance, through cross-selling offerings, cooperative advertising (Xin & Lin, 2024; Yan et al., 2023; Zhang et al., 2021) to ultimately increase their revenue. In addition, a coopetitive platform (Gulati et al., 2000) could alleviate the concerns regarding platform members' dependence on a single operator, such as Apple or Spotify (Lee, 2019). As platform owners and complementors engage in market integration (i.e., pool resources and join forces to develop and operate a unified music platform), they can share risks and overcome obstacles (e.g., lack of resources) to keep up with technology developments and compete with big digital platforms.

As reflected in DP3 and DP5, bundling limited add-on coopetitive subscriptions with unlimited base subscriptions and employing micropayments enable consumers to access and pay for their required or favourite content on demand. Reflecting on our results, we propose that offering consumers with such options could help attract users with various consumption needs and habits, which in turn could alleviate their reluctance to pay for online and copyrighted content such as news, music, and films. In addition, using trace data collected through the platform and utilizing analytics enable platform members to offer personalized content to consumers (Jenkins & Nielsen, 2020; Myllylahti, 2016; Puijk et al., 2021; Sjøvaag, 2016) and identify and attract potential future subscribers and ultimately increase their revenue.

Finally, as reflected in DP4, employing an SSO capability for the coopetitive platform allows readers to effortlessly access content from various producers and publishers in one place. This, coupled with integrating a micropayment feature or allowing users to top-up their accounts in the platform, could enhance the onboarding process for new subscribers and the user experience, which is shown to be an important factor for sustained use of a service or platform or user stickiness and loyalty (Liu & Wang, 2023; T.; Ajith et al., 2025).

5.2. Contributions to the coopetition and online newspaper literature

Our study offers several contributions to the literature. *First*, our findings contribute to the coopetitive literature on horizontal alliances (Hoffmann et al., 2018), in which coopetitive business network members operate in the same industry and compete for the same customers but have a joint industry-level problem that needs to be solved. Our study extends the literature on coopetitive horizontal alliances by offering an example of a successful coopetition venture, especially from the viewpoint of smaller online newspaper publishers, who are often reluctant to participate (Chiambaretto et al., 2020). Furthermore, our study offers an insider perspective of the collaboration dynamics (e.g., Gnyawali & Park, 2011) that are part of horizontal joint ventures.

The developed five DPs offer a novel contribution to the coopetition literature through a mid-range theorization of what is required for the design and business model of a coopetitive news platform—something the coopetition literature sees as challenging (Geurts et al., 2022). Furthermore, the literature often focuses on pre-competitive coopetition but not on delivering products or services to customers (Harala et al., 2023). In other words, horizontal R&D alliances (Hoffmann et al., 2018), as our technology and newspaper industry exemplars illustrate (see Section 2.2), are focal to this

research. Thus, we advance the coopetition literature by showing how to successfully design a news platform that enables publishers to pool resources and offerings (Bouncken et al., 2020; Pei, 2024) to cooperatively generate extra revenue through cross-selling and joint membership (Yan et al., 2023; Zhang et al., 2021) and, at the same time, deal with challenges such as a lack of resources and competition from big digital platforms. Thus, our DPs not only provide insights into how to organize and design a coopetitive news platform but also, more specifically, what characteristics are essential for enabling the cross-selling of products or services. So far, the literature on resolving this problem is scarce, and few studies (e.g., Luo et al., 2007) have investigated this.

Our findings also help practitioners develop and design similar news platforms, enabling small business owners to develop their digital offerings at a reasonable cost—something the recent coopetition literature has called for (Chiambaretto et al., 2020). We have summarized these findings for the newspaper industry in Table 8. We first distinguish the difference between our conceptualization (i.e., the abstract problem and solution) and practical design advice we offer for online newspaper publishers (i.e., the concrete problem and solution). The abstract problem relates to the problem we recognized from the literature and our discussions with the industry experts, namely how digitalization of newspapers has changed how we consume news. This led us to further to think how digital platforms has changed the business models of the newspaper and news delivery and what concrete problems these have created for the industry. Our DPs, in turn, offer concrete design solutions to address the problems. Finally, we abstract the offered solution to offer insights for other industry sectors that may benefit from developing coopetitive digital platforms, like the music industry as we discussed earlier.

Second, our study contributes to the IS literature at the intersection of online newspapers, coopetition and digital business models by proposing a novel add-on digital subscription model that can potentially facilitate a net expansion of the online news market (Calzada & Gil, 2020). By conceptualizing a coopetitive platform-based business model for cross-selling online news content, we concur with Harala et al. (2023), who argued for the need for a purposeful design of a platform-based business model to be successful in coopetition. The proposed new add-on subscription model, implemented as single and package plans, can be the basis for transforming online newspaper pricing, which has traditionally leaned towards offering fixed subscriptions. Our free simulation

Table 8. From abstract challenges to coopetitive digital platform solutions.

Abstract problem	Concrete problem in the newspaper industry	Concrete solution for the newspaper industry	Abstract Solution
Digitalization and the emergence of big digital platforms with a global reach have transformed traditional content distribution, enabling users to access and share free online content on a global scale.	Due to service fragmentation in the online newspaper industry, each newspaper must invest in developing its own digital news delivery channels or join monopolistic digital platforms. While competing with big digital platforms is challenging due to the lack of resources and capabilities, joining these platforms can lead to a loss of control over distribution and user relationships.	Forming a coopetition alliance between online newspaper publishers (DP3) for building a coopetitive digital news platform enables publishers to join efforts in and share risks of competing with big digital platforms (DP1) and engage in cross-selling online news by offering add-on news subscription plans (DP2). Such a coopetitive digital news platform enables publishers to expand their value network and market reach and co-create additional value propositions, which can ultimately increase the publishers' revenue.	A coopetitive digital platform for cross-selling online creative content enables content producers and publishers to co-create added value for consumers and join efforts in competing with big digital platforms.
	Due to service fragmentation in the online newspaper industry, each newspaper offers its own digital subscription to the readers. This increases adoption costs for readers since most users prefer to read news from multiple sources and would require multiple subscriptions. Such adoption costs and the availability of free news on the Internet reduce readers' willingness to pay for online news subscriptions.	Integrating micropayment (DP5) and single sign-on (SSO) capabilities (DP4) into the coopetitive digital news platform alongside offering add-on news subscription plans (DP2) provides readers with seamless and affordable access to news aggregated from various reliable sources. Offering such complementary value propositions and enhancing the user experience can help attract users with various consumption needs and habits, which can ultimately increase their willingness to pay for online news content.	

experiment and concept testing survey results show that there is a clear demand for the proposed plans.

Our results show that by offering such add-on plans and access to a larger pool of news content, publishers can extend their reach and attract those readers who do not consume news heavily (e.g., daily) or who would prefer to purchase valuable online news content on demand. By complementing fixed-term subscriptions with add-on subscriptions and joint memberships, news publishers can resolve the problems related to scaling up their digital subscriber base (Rios-Rodríguez et al., 2024) and convincing readers to pay for digital content (Chyi & Tenenboim, 2019; Sjøvaag & Owren, 2021), especially younger and low-income readers (Olsen & Solvoll, 2018). As such, we offer an example of how coopetition can create business model innovations without needing disruptive new technology innovations (Ritala & Sainio, 2014). The industry association also requested the source code for our IT artefact, which is also available as open source in GitHub,[10] indicating that they saw value in the project's outcome and our findings.

This finding is significant because the newspaper industry has been in a downward spiral regarding its market size (see, e.g., Winter & Alpar, 2018). Furthermore, while the literature offers guidance on monetizing online content (e.g., Pattabhiramaiah et al., 2019), small newspaper publishers have struggled to make their online news offerings profitable (Olsen & Solvoll, 2018). Berger (2018), for example, highlighted the monetization problems of newspaper publishers, while others have examined churn and conversion rates (Koch & Benlian, 2017; Winter & Alpar, 2018). Adding to these difficulties, an earlier steady source of revenue, online advertising, is declining (Jenkins & Nielsen, 2020; Rios-Rodríguez et al., 2024). Consequently, new business models and digitalization of newspapers are needed to change this trajectory (Jenkins & Nielsen, 2020; Myllylahti, 2016; Puijk et al., 2021; Sjøvaag, 2016) because simply increasing subscription prices will not work (Chyi & Tenenboim, 2019). Thus, our findings offer a novel way to "slice the cake" so that the individual pieces are larger than the original (i.e., the market size). This offers opportunities for the online newspaper industry to rejuvenate its business and organize it coopetitively. Consequently, our findings offer a possible way forwards to support digitalization of the newspaper industry and other industry fields that face similar problems, where large global corporations can provide digital services and compete with local and regional newspapers (Sjøvaag & Owren, 2021). Hence, our findings not only address one class of problems—how to design a coopetitive news platform—but also offer wide-ranging implications for the study of digital business (see, e.g., Silva et al., 2018) and the design of digital services (see, e.g., Tuunanen et al., 2023).

5.3. Contributions to DSR practice

We also see some clear benefits from using the ^{e}DSR methodology, since it offers a clear structure for developing a design knowledge contribution to the literature. For instance, reporting our study has been relatively easy, despite the complex nature of the project with 16 design echelons. This outcome encourages us to recommend that other researchers use the ^{e}DSR methodology. To that end, our study extends the DSR literature by demonstrating how a complex ^{e}DSR study can be organized, conducted, and reported. While other authors have embraced ^{e}DSR, they have often focused only on certain aspects of the ^{e}DSR methodology. For example, Beermann et al. (2024) offered an insightful perspective on opening the problem analysis echelon and how researchers can navigate this first design echelon. Hoess et al. (2024), in turn, offered a description of analysing the problem and defining the objectives of a solution and how to validate these in a legal compliance setting. Vial and Grange (2024) applied ^{e}DSR to conceptualize design knowledge development for a digital service.

We offer a mid-range theorization of the findings and a pathway to further development of design theory. Our study produced design knowledge in the form of five DPs to develop the basis for a design theory for designing coopetitive news platforms. We depicted 16 echelons and their interactions to organize a complex DSR study with an industry association and multiple newspaper publishers. Furthermore, we demonstrate how various techniques can be employed to carry out and evaluate different design echelons. For instance, our supplementary material[11] provides researchers with instructions on how to plan and conduct online experiments and concept testing in future studies. This research also exemplified the modular structure of an ^{e}DSR, and we reported individual design echelons at related scientific conferences during the project (Kazan & Tuunanen, 2020; Kazan et al., 2020, 2021). Our own experiences with the ^{e}DSR have been very positive. It helped us manage the complexities of running a project with a large number of design echelons and several industry collaborators and validate our artefact and later DPs during each design echelon. While our study offers an illustrative template, that worked well in our ^{e}DSR project, future research is needed to scrutinize our template and perhaps offer other alternatives that might work well for other types of projects.

Lastly, we see one unexpected DSR outcome for our project. Often, DPs evolve a great deal during a DSR project (see, e.g., Tuunanen & Peffers, 2018). New DPs are found, and the original DPs are significantly revised during the project. While we developed some preliminary DPs in the early phases of the study (see, e.g., Kazan et al., 2020, 2021), the DPs remained stable after they were formed in Echelon 7, with only one DP being revised. We did not expect this, and the findings contradict those of our earlier DSR projects using other DSR methodologies. One could, of course, argue that our DPs did change significantly from the ones reported in our published conference papers. However, taking a retrospective view, we argue that our DOs and DRs were developed through numerous design thinking and brainstorming workshops within the research team as well as frequent interactions with the newspaper industry representatives. Therefore, the DPs were built on solid foundations and justificatory knowledge gained from the literature and analysis of the collected empirical data. We are keen to see whether other researchers have similar experiences when applying ^{e}DSR to their projects.

5.4. Limitations

Our study has some limitations. First, our industry collaborators work in Finnish and Finland. Consequently, our findings may not directly translate to more widely spoken languages, such as Chinese, French, English, German, or Hindi. However, we evaluated the artefact in the US (the free simulation experiments) and in the UK and Germany (the concept testing survey). The findings from the free simulation experiment with English-speaking MTurk workers indicated that the news platform was appealing to them. Similarly, the concept testing survey with UK and German panel members concurred with the earlier results.

While our results from the free simulation experiments ($n = 175$) and the concept testing survey ($n = 415$) are robust, we also recognize that other validation approaches could have been considered. One such opportunity would have been to design a prototype of a coopetitive online news platform that incorporated live news material from our online newspaper publishers. This was our original research design when we started the project. However, COVID-19 disrupted these plans, and our research client, who had agreed to implement and integrate the prototype with their online newspaper and digital subscription service, told us that due to *force majeure*, they were unable to fulfil their part of the project. This was quite understandable and, in hindsight, gave us more freedom to evaluate the DPs. Still, conducting field experiments with the prototype with real online readers and the money they spend on digital subscriptions using our news platform would have been exciting.

6. Conclusions

Our study reports on a five-year ^{e}DSR project with the Finnish newspaper industry involving 16 design echelons. Our study contributes to the literature and practice by a mid-range design theory for designing coopetitive digital platforms that support cross-selling of online news content. We also offer a PoC and a PoV validation of the coopetitive news platform implemented in the project, which offers online newspaper publishers a robust base for adopting our findings and developing a coopetitive business in online newspapers and more broadly. We also showcase what a coopetitive business model can potentially do for their business; namely, there is a clear and distinct opportunity to "slice the cake" so that everyone, including online news readers, wins. In conclusion, we see great potential in the applied ^{e}DSR methodology. It offered a valuable and easy-to-follow structure for conducting a study and, later, a concise way to report our 16 design echelons and evaluate our DPs. We invite other researchers to adopt and try the ^{e}DSR. We would be especially keen to understand whether other researchers' results with DP development correspond with ours.

Notes

1. Gregor and Hevner (2013) define a mid-range design theory as a well-developed design theory about embedded phenomena. Following Merton (1968) we characterize our design theory as mid-range since we develop generalizable design knowledge that is abductively derived from the reference theories and our data collection and analysis explaining a limited set of phenomena.
2. https://aboutus.ft.com/press_release/openai
3. The abductive approach was developed by Charles Peirce. More about his work can be read at https://plato.stanford.edu/entries/peirce. Daase et al. (2024) recently reviewed deductive, inductive, and abductive approaches to theorizing based on design knowledge in the DSR literature.
4. For *Implementer I* to achieve or allow for *Aim A* for *User U* in *Context C*, employ *Mechanisms M1, M2,... Mn*, involving *Enactors E1, E2,... En because* of *Rationale R*.
5. The snapshots of the artefact are shown in Appendix S2, and the task description is reported in our supplementary material https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5273825.
6. See our supplementary material the compensation details and calculations https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5273825.
7. See our supplementary for the full survey https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5273825.

8. More specifically, we found that 61% of respondents aged 31–40 years and 51% of respondents with out an university degree, and earning earning 19500–65,000 € preferred the package plan. Interestingly, we also that 51% of respondents with an university degree with similar earnings and age also preferred the package plan.
9. Design theory building and echelon details are depicted in Appendix S1.
10. https://github.com/EmpGator/news-app
11. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5273825

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was funded by the Media Industry Research Foundation of Finland and Tietotekniikan Säätiö. The foundations have not participated in the preparation of this article.

References

Ajith, N. T., Sridevi, P., & Mathew, L. S. (2025). Experience matters: Exploring the impact of user experience on stickiness and loyalty in OTT platforms. *International Journal of human-Computer Interaction*, 1–15. https://doi.org/10.1080/10447318.2024.2443245

Al-Debei, M. M., & Avison, D. (2010). Developing a unified framework of the business model concept. *European Journal of Information Systems*, *19*(3), 359–376. https://doi.org/10.1057/ejis.2010.21

Amit, R., & Zott, C. (2001). Value creation in e-business. *Strategic Management Journal*, *22*(6–7), 493–520. https://doi.org/10.1002/smj.187

Balasubramanian, S., Bhattacharya, S., Krishnan, V. V. (2015). Pricing information goods: A strategic analysis of the selling and pay-per-use mechanisms. *Marketing Science*, *34*(2), 218–234. https://doi.org/10.1287/mksc.2014.0894

Beermann, V., Rieder, A., Uebernickel, F., & Vom Brocke, J. (2024). *Disentangling the problem space: A validated problem statement for sustainability support Systems*. International Conference on Design Science Research in Information Systems and Technology, Trollhättan, Sweden.

Berger, B. (2018). Commerce-oriented revenue models for content providers: An experimental study of commerciality's effect on credibility. *Electronic Markets*, *28*(1), 93–109. https://doi.org/10.1007/s12525-017-0268-z

Bhuller, M., Havnes, T., McCauley, J., & Mogstad, M. (2024). How the internet changed the market for print media. *American Economic Journal: Applied Economics*, *16*(2), 318–358. https://doi.org/10.1257/app.20210689

Bouncken, R. B., Fredrich, V., & Kraus, S. (2020). Configurations of firm-level value capture in coopetition. *Long Range Planning*, *53*(1), 101869. https://doi.org/10.1016/j.lrp.2019.02.002

Bouncken, R. B., Gast, J., Kraus, S., & Bogers, M. (2015). Coopetition: A systematic review, synthesis, and future research directions. *Review of Managerial Science*, *9*(3), 577–601. https://doi.org/10.1007/s11846-015-0168-6

Brandenburger, A., & Nalebuff, B. (2021). The rules of co-opetition. *Harvard Business Review*, *99*(1), 48–57.

Calzada, J., & Gil, R. (2020). What do news aggregators do? Evidence from Google news in Spain and Germany. *Marketing Science*, *39*(1), 134–167. https://doi.org/10.1287/mksc.2019.1150

Chandra, A., & Collard-Wexler, A. (2009). Mergers in two-sided markets: An application to the Canadian newspaper industry. *Journal of Economics & Management Strategy*, *18*(4), 1045–1070. https://doi.org/10.2139/ssrn.985581

Chen, Y.-J., & Seshadri, S. (2007). Product development and pricing strategy for information goods under heterogeneous outside opportunities. *Information Systems Research*, *18*(2), 150–172. https://doi.org/10.1287/isre.1070.0119

Chiambaretto, P., Bengtsson, M., Fernandez, A.-S., & Näsholm, M. H. (2020). Small and large firms' trade-off between benefits and risks when choosing a coopetitor for innovation. *Long Range Planning*, *53*(1), 101876. https://doi.org/10.1016/j.lrp.2019.03.002

Choi, P., Garcia, R., & Friedrich, C. (2010). The drivers for collective horizontal coopetition: A case study of screw-cap initiatives in the international wine industry. *International Journal of Strategic Business Alliances*, *1*(3), 271–290. https://doi.org/10.1504/IJSBA.2010.030427

Chyi, H. I., & Jeong, S. H. (2024). Unraveling US newspapers' digital and print subscriptions in the context of price, 2016-2022. *Media and Communication*, *12*, 1–20. https://doi.org/10.17645/mac.7482

Chyi, H. I., & Tenenboim, O. (2017). Reality check: Multiplatform newspaper readership in the United States, 2007-2015. *Journalism Practice*, *11*(7), 798–819. https://doi.org/10.1080/17512786.2016.1208056

Chyi, H. I., & Tenenboim, O. (2019). Charging more and wondering why readership declined? A longitudinal study of US newspapers' price hikes, 2008-2016. *Journalism Studies*, *20*(14), 2113–2129. https://doi.org/10.1080/1461670X.2019.1568903

Corbo, L., Kraus, S., Vlačić, B., Dabić, M., Caputo, A., & Pellegrini, M. M. (2023). Coopetition and innovation: A review and research agenda. *Technovation*, *122*, 102624. https://doi.org/10.1016/j.technovation.2022.102624

Daase, C., Haertel, C., Nahhas, A., & Turowski, K. (2024). *Classifying design Science research in terms of types of reasoning from an epistemological perspective*. International Conference on Design Science Research in Information Systems and Technology, Trollhättan, Sweden.

DiMaggio, P. J. (1995). Comments on"what theory is not". *Administrative Science Quarterly*, *40*(3), 391–397. https://doi.org/10.2307/2393790

Ellis, A. P., Hollenbeck, J. R., Ilgen, D. R., Porter, C. O., West, B. J., & Moon, H. (2003). Team learning: Collectively connecting the dots. *Journal of Applied Psychology*, *88*(5), 821–835. https://doi.org/10.1037/0021-9010.88.5.821

Eriksson, M., Fleischer, R., Johansson, A., Snickars, P., & Vonderau, P. (2019). *Spotify teardown: Inside the black box of streaming music*. MIT Press.

Evans, D. S. (2009). The online advertising industry: Economics, evolution, and privacy. *Journal of Economic Perspectives*, *23*(3), 37–60. https://doi.org/10.1257/jep.23.3.37

Fetscherin, M., & Knolmayer, G. (2004). Business models for content delivery: An empirical analysis of the newspaper and magazine industry. *The International Journal on Media Management*, *6*(1–2), 4–11. https://doi.org/10.1080/14241277.2004.9669377

Fishburn, P. C., Odlyzko, A. M., & Siders, R. C. (2000). Fixed fee versus unit pricing for information goods: Competition, equilibria, and price wars. In B. Kahin & H. R. Varian (Eds.), *Internet publishing and beyond: The economics of digital information and intellectual property* (pp. 167–189). The MIT Press.

Fromkin, H. L., & Streufert, S. (1976). Laboratory experimentation. In M. Dunnette (Ed.), *Handbook of organizational and industrial psychology* (pp. 415–465). Rand McNally.

Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, *27*(1), 51–90. https://doi.org/10.2307/30036519

Geurts, A., Broekhuizen, T., Dolfsma, W., & Cepa, K. (2022). Tensions in multilateral coopetition: Findings from the disrupted music industry. *Industrial Marketing Management*, *105*, 532–547. https://doi.org/10.1016/j.indmarman.2022.06.020

Gnyawali, D. R., & Park, B.-J. R. (2011). Co-opetition between giants: Collaboration with competitors for technological innovation. *Research Policy*, *40*(5), 650–663. https://doi.org/10.1016/j.respol.2011.01.009

Golovianko, M., Gryshko, S., Terziyan, V., & Tuunanen, T. (2023). Responsible cognitive digital clones as decision-makers: A design science research study. *European Journal of Information Systems*, *32*(5), 879–901. https://doi.org/10.1080/0960085X.2022.2073278

Gregor, S., Chandra Kruse, L., & Seidel, S. (2020). Research perspectives: The anatomy of a design principle. Association for Information Systems.

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, *37*(2), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01

Gulati, R., Nohria, N., & Zaheer, A. (2000). Strategic networks. *Strategic Management Journal*, *21*(3), 203–215. https://doi.org/10.1002/(SICI)1097-0266(200003)21:3<203::AID-SMJ102>3.0.CO;2-K

Gulati, R., Wohlgezogen, F., & Zhelyazkov, P. (2012). The two facets of collaboration: Cooperation and coordination in strategic alliances. *Academy of Management Annals*, *6*(1), 531–583. https://doi.org/10.5465/19416520.2012.691646

Handke, C. (2020). Music industry. In *Handbook of Cultural Economics, Third Edition* (pp. 358–370). Edward Elgar Publishing.

Hannah, D. P., & Eisenhardt, K. M. (2018). How firms navigate cooperation and competition in nascent ecosystems. *Strategic Management Journal*, *39*(12), 3163–3192. https://doi.org/10.1002/smj.2750

Harala, L., Aarikka-Stenroos, L., & Ritala, P. (2023). Coopetition for a circular economy: Horizontal initiatives in resolving collective environmental challenges. In J. Kujala, A. Heikkinen, & A. Blomberg (Eds.), *Stakeholder engagement in a sustainable circular economy: Theoretical and practical perspectives* (pp. 311–362). Springer International Publishing Cham.

Hedman, J., & Kalling, T. (2003). The business model concept: Theoretical underpinnings and empirical illustrations. *European Journal of Information Systems*, *12*(1), 49–59. https://doi.org/10.1057/palgrave.ejis.3000446

Hevner, A., March, S., Park, J., & Ram, S. (2004). Design Science in information Systems research. *MIS Quarterly*, *28*(1), 75. https://doi.org/10.2307/25148625

Hoess, A., Pocher, N., Roth, T. H., & Fridgen, G. (2024). *Towards a design Science research process for legal compliance by design*. Pacific-Asia Conference on Information Systems, Ho Chi Minh City, Vietnam.

Hoffmann, W., Lavie, D., Reuer, J. J., & Shipilov, A. (2018). The interplay of competition and cooperation. *Strategic Management Journal*, *39*(12), 3033–3052. https://doi.org/10.1002/smj.2965

Holm, A. B., Günzel, F., & Ulhøi, J. P. (2013). Openness in innovation and business models: Lessons from the newspaper industry. *International Journal of Technology Management*, *61*(3/4), 324–348. https://doi.org/10.1504/IJTM.2013.052674

Jääskeläinen, A., Yanatma, S., & Ritala, P. (2021). How does an incumbent news media organization become a platform? Employing intra-firm synergies to launch the platform business model in a news agency. *Journalism Studies*, *22*(15), 2061–2081. https://doi.org/10.1080/1461670X.2021.1979426

Jenkins, J., & Nielsen, R. K. (2020). Preservation and evolution: Local newspapers as ambidextrous organizations. *Journalism*, *21*(4), 472–488. https://doi.org/10.1177/1464884919886421

Jensen, F. J. (2024). Rethinking royalties: Alternative payment systems on music streaming platforms. *Journal of Cultural Economics*, *48*(3), 439–462.

Karimi, J., & Walter, Z. (2015). The role of dynamic capabilities in responding to digital disruption: A factor-based study of the newspaper industry. *Journal of Management Information*, *32*(1), 39–81. https://doi.org/10.1080/07421222.2015.1029380

Karimi, J., & Walter, Z. (2016). Corporate entrepreneurship, disruptive business model innovation adoption, and its performance: The case of the newspaper industry. *Long Range Planning*, *49*(3), 342–360. https://doi.org/10.1016/j.lrp.2015.09.004

Kazan, E., Tan, C.-W., Lim, E. T., Sørensen, C., & Damsgaard, J. (2018). Disentangling digital platform competition: The case of UK mobile payment platforms. *Journal of Management Information*, *35*(1), 180–219. https://doi.org/10.1080/07421222.2018.1440772

Kazan, E., & Tuunanen, T. (2020, December 2–4). *The design towards a collaborative subscription service: The case of the Finnish newspaper industry*. Proceedings of the 15th International Conference on Design Science Research in Information Systems and Technology, DESRIST 2020, Kristiansand, Norway.

Kazan, E., Tuunanen, T., Ghanbari, H., & Li, M. (2021). *The design towards a news aggregator subscription service: Results from an online experiment*. Proceedings of the Annual Hawaii International Conference on System Sciences, Hawaii, USA.

Kazan, E., Tuunanen, T., Li, M., Ghanbari, H., & Tumbas, S. (2020). Strategic design towards platform collaboration in the newspaper industry: A design science research study.

Klus, M. F., Lohwasser, T. S., Holotiuk, F., & Moormann, J. (2019). Strategic alliances between banks and fintechs for digital innovation: Motives to collaborate and types of interaction. *The Journal of Entrepreneurial Finance (JEF)*, *21*(1), 1–23. https://doi.org/10.57229/2373-1761.1346

Koch, O. F., & Benlian, A. (2017). The effect of free sampling strategies on freemium conversion rates. *Electronic*

Markets, *27*(1), 67–76. https://doi.org/10.1007/s12525-016-0236-z

Kraus, S., Schmid, J., & Gast, J. (2017). Innovation through coopetition: An analysis of small-and medium-sized trust companies operating in the Liechtenstein financial centre. *International Journal of Business Science & Applied Management*, *12*(1), 44–60. https://doi.org/10.69864/ijbsam.12-1.124

Lee, E. (2019). Media companies take a big gamble on Apple. June 7.

Le Roy, F., & Czakon, W. (2016). Managing coopetition: The missing link between strategy and performance. *Industrial Marketing Management*, *53*(1), 3–6. https://doi.org/10.1016/j.indmarman.2015.11.005

Li, S., Sun, B., & Montgomery, A. L. (2011). Cross-selling the right product to the right customer at the right time. *Journal of Marketing Research*, *48*(4), 683–700. https://doi.org/10.1509/jmkr.48.4.683

Liu, Y., & Wang, Y. (2023). Empirical study on the factors affecting user stickiness of online visual art platform from the perspective of user experience. *Institute of Electrical and Electronics Engineers Access*, *11*, 60763–60776. https://doi.org/10.1109/ACCESS.2023.3284541

Luo, X., Rindfleisch, A., & Tse, D. K. (2007). Working with rivals: The impact of competitor alliances on financial performance. *Journal of Marketing Research*, *44*(1), 73–83. https://doi.org/10.1509/jmkr.44.1.73

Magretta, J. (2002). Why business models matter. *Harvard Business Review*, *80*(5), 86–92, 133.

Malms, O., & Schmitz, C. (2011). Cross-divisional orientation: Antecedents and effects on cross-selling success. *Journal of business-To-Business Marketing*, *18*(3), 253–275. https://doi.org/10.1080/1051712X.2011.541377

Mathias, B. D., Huyghe, A., Frid, C. J., & Galloway, T. L. (2018). An identity perspective on coopetition in the craft beer industry. *Strategic Management Journal*, *39*(12), 3086–3115. https://doi.org/10.1002/smj.2734

Meena, A., Dhir, S., & Sushil, S. (2023). A review of coopetition and future research agenda. *Journal of Business & Industrial Marketing*, *38*(1), 118–136. https://doi.org/10.1108/JBIM-09-2021-0414

Merton, R. K. (1968). *Social theory and social structure*. Simon and Schuster.

Myllylahti, M. (2016). Newspaper paywalls and corporate revenues: A comparative study. In S. A. Eldridge II & B. Franklin (Eds.), *The Routledge companion to digital journalism studies* (pp. 166–175). Routledge.

Nalebuff, B. J., & Brandenburger, A. M. (1997). Co-opetition : Competitive and cooperative business strategies for the digital economy. *Strategy & Leadership*, *25*(6), 28–33. https://doi.org/10.1108/eb054655

Nguyen, A., Tuunanen, T., Gardner, L., & Sheridan, D. (2021). Design principles for learning analytics information systems in higher education. *European Journal of Information Systems*, *30*(5), 541–568. https://doi.org/10.1080/0960085X.2020.1816144

Nunes, J. C. (2000). A cognitive model of people's usage estimations. *Journal of Marketing Research*, *37*(4), 397–409. https://doi.org/10.1509/jmkr.37.4.397.18788

Olsen, R. K., & Solvoll, M. K. (2018). Bouncing off the paywall-understanding misalignments between local newspaper value propositions and audience responses. *The International Journal on Media Management*, *20*(3), 174–192. https://doi.org/10.1080/14241277.2018.1529672

Osterwalder, A., Pigneur, Y., & Tucci, C. L. (2005). Clarifying business models: Origins, present, and future of the concept. *Communications of the Association for Information Systems*, *16*(1), 1–25. https://doi.org/10.17705/1CAIS.01601

Ozer, M. (1999). A survey of new product evaluation models. *Journal of Product Innovation Management: An International Publication of the Product Development & Management Association*, *16*(1), 77–94. https://doi.org/10.1111/1540-5885.1610077

Pang, M.-S., & Etzion, H. (2012). Research note-analyzing pricing strategies for online services with network effects. *Information Systems Research*, *23*(4), 1364–1377. https://doi.org/10.1287/isre.1110.0414

Pattabhiramaiah, A., Sriram, S., & Manchanda, P. (2019). Paywalls: Monetizing online content. *Journal of Marketing*, *83*(2), 19–36. https://doi.org/10.1177/0022242918815163

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information*, *24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Pei, L. (2024). Winner-takes-all or competitive coexistence? Research on the co-opetition relationships between platform owners and complementors. *Chinese Management Studies*, *18*(6), 1678–1707. https://doi.org/10.1108/CMS-03-2023-0102

Peng, L., & Finn, A. (2008). Concept testing: The state of contemporary practice. *Marketing Intelligence & Planning*, *26*(6), 649–674. https://doi.org/10.1108/02634500810902884

Picard, R. G. (2018). The Economics of journalism and news provision. In T. P. Vos (Ed.), *Handbooks of communication science* (pp. 281–296). De Gruyter Mouton.

Piepponen, A., Ritala, P., Keränen, J., & Maijanen, P. (2022). Digital transformation of the value proposition: A single case study in the media industry. *Journal of Business Research*, *150*, 311–325. https://doi.org/10.1016/j.jbusres.2022.05.017

Poell, T. (2020). Three challenges for media studies in the age of platforms. *Television & New Media*, *21*(6), 650–657.

Puijk, R., Hestnes, E. B., Holm, S., Jakobsen, A., & Myrdal, M. (2021). Local newspapers' transition to online publishing and video use: Experiences from Norway. *Journalism Studies*, *22*(9), 1123–1141. https://doi.org/10.1080/1461670X.2021.1922303

Reid, S. E., & Crawford, C. B. (2022). *Concept testing (entrepreneurial vision: A guide for charting and implementing the visioning process*. Springer.

Rios-Rodríguez, R., Fernández-López, S., Dios-Vicente, A., & Rodeiro-Pazos, D. (2024). Digital opportunities for local journalism: A panel data analysis on the economic performance of online-only vs. *Digital Journalism*, *12*(1), 63–82. https://doi.org/10.1080/21670811.2022.2103009

Ritala, P. (2012). Coopetition strategy-when is it successful? Empirical evidence on innovation and market performance. *British Journal of Management*, *23*(3), 307–324. https://doi.org/10.1111/j.1467-8551.2011.00741.x

Ritala, P., & Hurmelinna-Laukkanen, P. (2009). What's in it for me? Creating and appropriating value in innovation-related coopetition. *Technovation*, *29*(12), 819–828. https://doi.org/10.1016/j.technovation.2009.07.002

Ritala, P., & Sainio, L.-M. (2014). Coopetition for radical innovation: Technology, market and business-model perspectives. *Technology Analysis & Strategic Management*, *26*(2), 155–169. https://doi.org/10.1080/09537325.2013.850476

Ryan-Charleton, T., Gnyawali, D. R., & Oliveira, N. (2022). Strategic alliance outcomes: Consolidation and new directions. *Academy of Management Annals*, *16*(2), 719–758. https://doi.org/10.5465/annals.2020.0346

Sætre, A. S., & Van de Ven, A. (2021). Generating theory by abduction. *Academy of Management Review*, *46*(4), 684–701. https://doi.org/10.5465/amr.2019.0233

Silva, B. N., Khan, M., & Han, K. (2018). Towards sustainable smart cities: A review of trends, architectures, components, and open challenges in smart cities. *Sustainable Cities and Society*, *38*, 697–713. https://doi.org/10.1016/j.scs.2018.01.053

Sjøvaag, H. (2016). Introducing the paywall: A case study of content changes in three online newspapers. *Journalism Practice*, *10*(3), 304–322. https://doi.org/10.1080/17512786.2015.1017595

Sjøvaag, H., & Owren, T. (2021). The non-substitutability of local news? Advertising and the decline of journalism's umbrella market model. *Nordicom Review*, *42*(1), 1–14. https://doi.org/10.2478/nor-2021-0001

Söllner, M., Hoffmann, A., & Leimeister, J. M. (2016). Why different trust relationships matter for information systems users. *European Journal of Information Systems*, *25*(3), 274–287. https://doi.org/10.1057/ejis.2015.17

Stadtler, L., & Van Wassenhove, L. N. (2016). Coopetition as a paradox: Integrative approaches in a multi-company, cross-sector partnership. *Organization Studies*, *37*(5), 655–685. https://doi.org/10.1177/0170840615622066

Streufert, S., Streufert, S. C., & Castore, C. H. (1968). Leadership in negotiations and the complexity of conceptual structure. *Journal of Applied Psychology*, *52*(3), 218–223. https://doi.org/10.1037/h0025852

Sundararajan, A. (2004). Nonlinear pricing of information goods. *Management Science*, *50*(12), 1660–1673. https://doi.org/10.1287/mnsc.1040.0291

Thorén, C., Ågerfalk, P. J., & Edenius, M. (2014). Through the printing press: An account of open practices in the Swedish newspaper industry. *Journal of the Association for Information Systems*, *15*(11), 779–804. https://doi.org/10.17705/1jais.00379

Tilson, D., Sørensen, C., & Lyytinen, K. (2013, January). Platform complexity: Lessons from the music industry. 46th Hawaii International Conference on System Sciences. (pp. 4625–4634). IEEE.

Timmers, P. (1998). Business models for electronic markets. *Electronic Markets*, *8*(2), 3–8. https://doi.org/10.1080/10196789800000016

Tofalvy, T., & Koltai, J. (2023). "Splendid isolation": The reproduction of music industry inequalities in Spotify's recommendation system. *New Media and Society*, *25*(7), 1580–1604.

Tuunanen, T., Lumivalo, J., Vartiainen, T., Zhang, Y., & Myers, M. D. (2023). Micro-level mechanisms to support value Co-creation for design of digital services. *Journal of Service Research*, *27*(3), 381–396. https://doi.org/10.1177/10946705231173116

Tuunanen, T., & Peffers, K. (2018). Population targeted requirements acquisition. *European Journal of Information Systems*, *27*(6), 686–711. https://doi.org/10.1080/0960085X.2018.1476015

Tuunanen, T., Winter, R., & Vom Brocke, J. (2024). Dealing with complexity in design Science research: Using design echelons to support Planning, conducting, and communicating design knowledge contributions. *MIS Quarterly*, *48*(2), 427–458. https://doi.org/10.25300/MISQ/2023/16700

Vial, G., & Grange, C. (2024). Conceptualizing digital service: Coconstitutive essence and value cocreation dynamics. *Journal of Service Management*, *35*(3), 408–437. https://doi.org/10.1108/JOSM-12-2023-0520

Viglia, G., Maras, M., Schumann, J., & Navarro-Martinez, D. (2019). Paying before or paying after? Timing and uncertainty in pay-what-you-want pricing. *Journal of Service Research*, *22*(3), 272–284. https://doi.org/10.1177/1094670519835308

Waldfogel, J. (2012). Digitization and copyright: Some recent evidence from music. *Communications of the ACM*, *55*(5), 35–37.

Winter, P., & Alpar, P. (2018). On the relationship between print and mobile channels for newspapers. *Electronic Markets*, *28*(1), 79–92. https://doi.org/10.1007/s12525-017-0263-4

Xin, W., & Lin, J. (2024). Early entrant's strategy for cooperating with a competing fourth-party platform. *rairo-Operations Research*, *58*(4), 2991–3018. https://doi.org/10.1051/ro/2024101

Yadav, N., Kumar, R., & Malik, A. (2022). Global developments in coopetition research: A bibliometric analysis of research articles published between 2010 and 2020. *Journal of Business Research*, *145*, 495–508. https://doi.org/10.1016/j.jbusres.2022.03.005

Yan, K., Hua, G., Cheng, T., Choi, T.-M., Dong, J.-X., & Li, X. (2023). Optimal pricing and quality decisions under cooperative promotion of cross-market service platforms. *IEEE Transactions on Engineering Management*, 10091–10115. https://doi.org/10.1109/TEM.2023.3301886

Yoo, Y., Henfridsson, O., & Lyytinen, K. (2010). Research commentary-the new organizing logic of digital innovation: An agenda for information systems research. *Information Systems Research*, *21*(4), 724–735. https://doi.org/10.1287/isre.1100.0322

Zhang, L., Chen, F.-W., Xia, S.-M., Cao, D.-M., Ye, Z., Shen, C.-R., Maas, G., & Li, Y.-M. (2021). Value co-creation and appropriation of platform-based alliances in cooperative advertising. *Industrial Marketing Management*, *96*, 213–225. https://doi.org/10.1016/j.indmarman.2021.06.001